

SOFTWARE DESIGN DOCUMENT

Risk-Based Internal Audit System (RBIAS)

A Comprehensive Web-Based Platform for Bank Branch Internal Audit

Planning • Execution • Reporting • Compliance Tracking

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Table of Contents

Table of Contents	2
Executive Summary	6
System Overview	8
System Context	8
User Roles and Personas	8
System Modules	9
System Architecture	11
Technology Stack	11
Architecture Pattern	11
API Design	12
Database Schema	13
Core Entity Tables	13
branches	13
audits	14
audit_team_members	15
audit_sections	15
audit_observations	17
loan_reviews	17
sma_npa_entries	18
cash_checks	20
RAM and Planning Tables	20
ram_assessments	20
ram_parameters (Reference/Config)	21
Compliance Tracking Tables	22
compliance_items	22
Value Statement / Examination Master Tables	23
examination_areas	23
examination_items	23
audit_examination_responses	24
Supporting Tables	26
Entity Relationship Summary	27
User Interface Design	28
Screen Inventory	28
Key UI Patterns	30
Audit Execution Interface	30

Observation Entry Pattern	30
Compliance Tracker UI.....	31
Dashboard Wireframe Specifications	31
Feature Specifications	32
M1: Audit Planning Module	32
RAM Computation Engine	32
Annual Audit Plan Generator.....	33
M3: Audit Execution Module	34
Value Statement Examination Framework	34
Examination Workflow per Functional Area.....	35
Credit / Advances: Deep-Dive Examination	36
Section Data Structures.....	37
Smart Features for Field Auditors.....	37
M4: Report Generation Module.....	39
Risk Rating Computation.....	39
M5: Compliance Tracking Module.....	40
Compliance Workflow	40
Escalation Rules	40
M6: Analytics & Dashboards Module	41
Key Analytics Views	41
M8: Enterprise Risk Register & Audit Universe	42
Audit Universe	42
Enterprise Risk Register	42
Enhanced Risk-Based Plan Inputs.....	43
M9: Control Library & Standardized Work Programs.....	44
Control Library.....	44
Test Procedures & Work Programs	44
Control Effectiveness Analytics	45
M10: Continuous Auditing & Data-Driven Testing	46
Data Integration Layer.....	46
Rule-Based Analytics Engine.....	46
Data Exceptions & Continuous Monitoring	47
M11: AI & Advanced Analytics Layer.....	48
Anomaly Detection	48
Text Analytics & Predictive Models	48
Smart Suggestions UI	48

M12: Issue & Action Management.....	49
Issues Table.....	49
Action Plans	49
Enhanced Planning & Resource Management.....	51
Auditor Skills & Capacity	51
Geography-Aware Scheduling	51
Multi-Entity Audits	51
GRC Integration & Open API Layer.....	52
Integration Points	52
Shared Taxonomies & Unified Risk View.....	52
UX & Mobility Enhancements	53
Mobile Application / PWA.....	53
Electronic Workpaper Management.....	53
Collaboration Features.....	53
Multi-Tenant Architecture.....	54
Tenant Isolation & Configuration.....	54
Deployment Models	54
M13: Quality Management for Internal Audit Function	55
Quality Assessment Framework	55
Internal Audit Effectiveness KPIs	55
UCB & RBI Regulatory Alignment Framework	57
M14: Unified Audit Universe & Calendar.....	58
Audit Universe Entity Table	58
M15: Concurrent Audit Module	60
Concurrent Audit Scope Templates.....	60
Concurrent Audit Workbench	60
M16: IRAC & Provisioning Engine	61
IRAC Computation Pipeline.....	61
M17: Investment & Treasury Audit Controls	62
M18: EDP / IS Audit Module	63
Application Inventory	63
IS Audit Checklists.....	63
M19: Regulatory Observation & Follow-up Hub.....	64
Housekeeping Risk Metrics.....	65
Inter-bank Exposure Monitoring	66
M20: Governance & Board Compliance Module.....	67

ACB Workspace & Agenda Builder	67
Board Review Calendar.....	67
Policy Library & Review Tracker.....	68
Committee Governance Metadata.....	68
RBI Inspection Support Pack	69
Risk Management MIS for RMC / Board	70
Non-Functional Requirements	71
Implementation Roadmap.....	73
Phase 1: Foundation (Weeks 1-6)	73
Phase 2: Audit Execution (Weeks 7-14)	73
Phase 3: Reporting & Compliance (Weeks 15-20)	73
Phase 4: Planning & Analytics (Weeks 21-26).....	73
Phase 5: Advanced Analytics & Continuous Auditing (Weeks 27-36)	74
Phase 6: GRC Integration & Issue Management 2.0 (Weeks 37-46)	74
Phase 7: Mobility, UX & Productization (Weeks 47-52+).....	74
Phase 8: UCB RBIA Core & Governance (Weeks 53-62).....	75
Phase 9: IRAC, Treasury, EDP & Risk Analytics (Weeks 63-72)	75
Appendix.....	77
A. Glossary.....	77
B. Reference Documents	78
C. Data Migration Strategy.....	79

Executive Summary

This Software Design Document (SDD) presents the complete architecture and specifications for the Risk-Based Internal Audit System (RBIAS), a web-based platform designed to digitize and streamline the end-to-end internal audit process for Urban Cooperative Banks (UCBs) and banking institutions operating under RBI regulatory supervision.

RBIAS is designed in strict alignment with (a) the Risk-Based Internal Audit (RBIA) Policy 2018, (b) RBI's Master Circular on Inspection & Audit Systems in UCBs, (c) IRAC norms for Income Recognition, Asset Classification and Provisioning, and (d) RBI directives on Concurrent Audit, EDP/IS Audit, Investment & Treasury controls, Capital Adequacy (CRAR), and Governance standards for UCBs. The system replaces the current Excel-based audit workflow with an integrated digital platform that supports all stakeholders — from field auditors and concurrent auditors to the Audit Committee of the Board (ACB) and Risk Management Committee (RMC).

Key Objectives:

1. Digitize the standardized 13-16 section audit report format currently maintained in Excel workbooks, enabling structured data capture with built-in validations, risk scoring, and real-time calculations. The system incorporates the complete IA Format (Internal Audit Examination Format) with its 12 functional areas and 220+ value statements as the standardized checklist driving audit execution.
2. Automate the Risk Assessment Model (RAM) for computing composite branch risk scores across 19 parameters, determining audit frequency, and generating the Annual Audit Plan.
3. Implement the full compliance lifecycle from initial audit observation through Branch Head certification, Zonal Audit Committee (ZAC) review, Audit Compliance and Evaluation (ACE) department processing, to final closure by the Audit Committee of the Board (ACB).
4. Provide role-based dashboards with real-time analytics for audit management, enabling trend analysis across branches, visits, and audit periods.
5. Generate standardized Excel/PDF audit reports matching the existing bank format for regulatory submissions and record-keeping.
6. Provide an enterprise-grade platform with continuous auditing via data connectors and analytics engine, AI-powered anomaly detection and smart suggestions, unified issue management across all assurance sources, a centralized control library with standardized work programs, and multi-tenant architecture for commercial deployment across multiple banks.
7. Deliver full UCB regulatory alignment — concurrent audit workbench, IRAC & provisioning engine, investment/treasury audit controls, EDP/IS audit module, inter-bank exposure monitoring, housekeeping KPIs, RBI observation follow-up hub, and Board/ACB/RMC governance dashboards — enabling a UCB to demonstrate comprehensive compliance during RBI inspections.

Technology Stack: React.js frontend, Node.js/Express backend, PostgreSQL database with RLS, deployed as a responsive web application with PWA mobile support accessible from laptops and tablets during branch visits. AI/ML layer for anomaly detection and text analytics.

System Overview

System Context

RBIAS operates within the bank's internal audit ecosystem, interfacing with the Inspection and Audit Department (IAD), branch operations, zonal offices, and the Board-level audit committee. The system serves as the central platform for all audit-related workflows.

User Roles and Personas

Role	Description	Key Responsibilities	Access Level
Field Auditor	Audit team members conducting branch inspections	Data capture, observation entry, risk rating, evidence attachment	Assigned audits only
Lead Auditor	Team leader for a specific audit engagement	Review team entries, finalize findings, compute ratings, submit report	Full audit management
Branch Head	Manager of the audited branch	Review findings, submit compliance responses, upload evidence	Own branch audits only
Zonal Auditor	ZAC member reviewing regional audit compliance	Track compliance status, escalate overdue items, approve closures	Zone-level oversight
IAD Manager	Inspection & Audit Department management	Annual audit planning, team assignment, RAM computation, policy config	Full system administration
ACE Officer	Audit Compliance & Evaluation department staff	Monitor compliance timelines, prepare ACB reports, manage escalations	All audit compliance data
ACB Member	Audit Committee of the Board	Review quarterly reports, approve policy changes, oversight dashboards	Read-only executive view
Concurrent Auditor	External CA firm or internal concurrent audit staff	Daily/weekly checks on designated branches and HO functions, rapid observation entry, serious irregularity escalation	Assigned concurrent audit entities only
IS/EDP Auditor	IT audit specialist (internal or outsourced)	IS audit of CBS, channels, access controls, vendor management, DR compliance	IS audit engagements and application inventory
Risk Head / RMC Member	Chief Risk Officer or Risk Management Committee member	Risk dashboards, KRI monitoring, capital adequacy tracking, inter-bank exposure oversight	Read-only risk management views
System Admin	Technical administrator	User management, system config, master data maintenance	Full system configuration

System Modules

RBIAS is organized into seven core modules, each addressing a distinct phase of the audit lifecycle:

Module	Purpose	Key Features
M1: Audit Planning	Annual audit plan creation and management	RAM computation, branch risk scoring, audit frequency determination, team assignment, calendar management
M2: Pre-Audit	Audit preparation and requisites	Branch data requisition, previous audit review, checklist preparation, document collection
M3: Audit Execution	Field audit data capture and observation entry	13-section structured data entry, risk categorization, evidence upload, offline capability, auto-calculations
M4: Report Generation	Standardized audit report creation	Excel/PDF export matching existing format, Branch Head certificate, summary generation, risk rating computation
M5: Compliance Tracking	Post-audit compliance monitoring	Branch response tracking, timeline management, escalation workflows, ZAC/ACE/ACB integration
M6: Analytics & Dashboards	Real-time audit intelligence	Branch scorecards, trend analysis, heat maps, KPI tracking, management reporting
M7: Administration	System configuration and master data	User/role management, branch master, audit parameters, policy configuration, notification templates
M8: Enterprise Risk Register	Audit universe and risk register	Auditable entities, risk statements, KRIs, risk-plan linkage, what-if simulation
M9: Control Library	Control testing and work programs	Control library, test procedures, work programs, control effectiveness analytics
M10: Continuous Auditing	Data-driven testing and monitoring	Data connectors/ETL, rule-based analytics, exception management, continuous monitoring dashboards
M11: AI Analytics	Machine learning and predictive insights	Anomaly detection, NLP text analytics, predictive risk models, smart suggestions
M12: Issue Management	Unified issue and action tracking	Cross-source issues, action plans, milestones, partial closure, consolidated Board reporting
M13: Quality Management	Audit function quality assurance	Self-assessments, gap analysis, effectiveness KPIs, external QA support
M14: Unified Audit Universe	All mandated audit types in one calendar	RBIA, internal inspection, concurrent, IS/EDP audits; surprise scheduling; RBI periodicity compliance
M15: Concurrent Audit	Early-warning concurrent audit workbench	RBI-aligned scope templates, rapid entry UI, serious irregularity escalation, de-duplication with RBIA

M16: IRAC Engine	NPA classification & provisioning cross-check	CBS data ingestion, IRAC recomputation, deviation detection, auto-exception generation
M17: Treasury Audit	Investment & treasury audit controls	SGL/CSGL reconciliation, broker compliance, non-SLR caps, quarterly certification workflow
M18: IS/EDP Audit	Technology & cyber audit	Application inventory, IS audit checklists, vendor risk, DR drill tracking
M19: Regulatory Hub	RBI/statutory observation follow-up	Para-to-issue mapping, ATR workflows, multi-source observation tracking
M20: Governance	Board/ACB/RMC compliance support	ACB workspace, agenda builder, Board review calendar, policy library, committee governance

System Architecture

Technology Stack

Layer	Technology	Rationale
Frontend	React.js 18+ with TypeScript, Tailwind CSS, shadcn/ui	Component-based UI with type safety, responsive design for tablet use during branch visits
State Management	Redux Toolkit with RTK Query	Centralized state for complex audit forms, built-in caching and API integration
Backend API	Node.js with Express.js, TypeScript	High-performance REST API with strong typing, excellent ecosystem for Excel/PDF generation
Database	PostgreSQL 15+	ACID compliance critical for audit data integrity, JSONB for flexible observation storage, full-text search
ORM	Prisma	Type-safe database access, automated migrations, excellent PostgreSQL support
Authentication	JWT with refresh tokens, RBAC middleware	Stateless auth suitable for distributed deployment, fine-grained role-based access control
File Storage	MinIO (S3-compatible) or local filesystem	Self-hosted object storage for audit evidence, documents, and generated reports
Report Engine	ExcelJS for XLSX, PDFKit for PDF generation	Pixel-perfect reproduction of existing bank audit report format
Search	PostgreSQL full-text search (pg_trgm)	Search across observations, borrower names, account numbers without additional infrastructure
Notifications	Nodemailer + in-app notification system	Email alerts for compliance deadlines, assignment notifications, escalation triggers
Deployment	Docker containers with Docker Compose	Consistent deployment across environments, easy scaling, isolated services

Architecture Pattern

RBIAS follows a layered architecture with clear separation of concerns:

Presentation Layer: React SPA with role-based routing. Each audit section (Cash Check, Loans & Advances, SMA & NPA, etc.) is a self-contained form module with local validation, auto-save, and progress tracking.

API Gateway Layer: Express.js REST API with JWT authentication middleware, request validation (Zod schemas), rate limiting, and audit logging. All API endpoints follow RESTful conventions with versioning (/api/v1/).

Business Logic Layer: Service classes encapsulating core business rules — RAM score computation, risk rating algorithms, compliance timeline calculations, report generation logic, and escalation rules as defined in the RBIA Policy.

Data Access Layer: Prisma ORM with repository pattern for database operations. All audit data changes are tracked with full audit trails (who changed what, when).

Infrastructure Layer: PostgreSQL for structured data, MinIO for file storage, Redis for session caching and job queues (Bull.js for background report generation).

API Design

The REST API is organized by module with consistent endpoint patterns:

Endpoint Group	Methods	Description
/api/v1/audits	GET, POST, PUT, DELETE	Audit engagement CRUD, status management, assignment
/api/v1/audits/:id/sections/:section	GET, PUT, PATCH	Section-level data entry (cash-check, loans, sma-npa, etc.)
/api/v1/audits/:id/observations	GET, POST, PUT, DELETE	Individual audit observations with risk categorization
/api/v1/audits/:id/report	GET, POST	Report generation and export (XLSX, PDF)
/api/v1/compliance	GET, PUT	Compliance tracking, branch responses, closure requests
/api/v1/planning/ram	GET, POST	RAM score computation and annual audit plan
/api/v1/branches	GET, POST, PUT	Branch master data management
/api/v1/users	GET, POST, PUT, DELETE	User management with role assignment
/api/v1/examination-areas	GET	Master list of 12 functional areas with item counts
/api/v1/examination-items	GET	Value statements master list, filterable by area
/api/v1/audits/:id/examinations	GET, PUT, PATCH	Audit examination responses for all value statements
/api/v1/audits/:id/examinations/:area	GET, PUT	Area-specific examination responses
/api/v1/analytics	GET	Dashboard data, KPIs, trend analytics
/api/v1/notifications	GET, PUT	Notification management and preferences

Database Schema

The database design captures the complete audit lifecycle from planning through compliance closure. All tables include standard audit columns (created_at, updated_at, created_by, updated_by) and soft-delete support (deleted_at).

Core Entity Tables

branches

Master table for all bank branches subject to audit.

Column	Type	Constraints	Description
id	UUID	PK, DEFAULT gen_random_uuid()	Unique branch identifier
branch_code	VARCHAR(10)	UNIQUE, NOT NULL	Bank's internal branch code (e.g., 0001)
branch_name	VARCHAR(100)	NOT NULL	Official branch name (e.g., Fort Main)
zone	VARCHAR(50)	NOT NULL, FK → zones.id	Zonal classification for ZAC reporting
region	VARCHAR(50)		Regional grouping
city	VARCHAR(50)	NOT NULL	City location
address	TEXT		Full branch address
branch_head_id	UUID	FK → users.id	Current Branch Head/Manager
branch_category	VARCHAR(20)	NOT NULL	Category: Large/Medium/Small/Very Small
business_size_lakhs	DECIMAL(15,2)		Total business in Lakhs for RAM computation
staff_strength	INTEGER		Number of staff at branch
is_currency_chest	BOOLEAN	DEFAULT false	Whether branch has currency chest
is_forex_branch	BOOLEAN	DEFAULT false	Whether branch handles forex transactions
last_audit_date	DATE		Date of most recent completed audit
last_audit_rating	VARCHAR(20)		Rating from last audit (Satisfactory/Needs Improvement/etc.)
ram_composite_score	DECIMAL(5,2)		Latest RAM composite risk score

audit_frequency_months	INTEGER		Derived audit frequency from RAM score
status	VARCHAR(20)	DEFAULT 'active'	active/inactive/merged

audits

Central table for each audit engagement. One record per audit visit to a branch.

Column	Type	Constraints	Description
id	UUID	PK	Unique audit identifier
audit_number	VARCHAR(20)	UNIQUE, NOT NULL	Sequential audit reference (e.g., RBIA/2024-25/001)
branch_id	UUID	FK → branches.id, NOT NULL	Branch being audited
audit_type	VARCHAR(30)	NOT NULL	Type: risk_based / snap / follow_up / concurrent / special
visit_number	INTEGER	DEFAULT 1	1st visit, 2nd visit, etc. for same audit period
audit_period_from	DATE	NOT NULL	Start of period under audit (e.g., Dec 2023)
audit_period_to	DATE	NOT NULL	End of period under audit (e.g., Aug 2024)
planned_start_date	DATE		Scheduled audit commencement date
actual_start_date	DATE		Actual date audit team arrived at branch
planned_end_date	DATE		Scheduled completion date
actual_end_date	DATE		Actual completion date
lead_auditor_id	UUID	FK → users.id	Lead auditor assigned to this engagement
status	VARCHAR(30)	NOT NULL, DEFAULT 'planned'	planned / in_progress / draft / submitted / reviewed / closed
overall_risk_rating	VARCHAR(30)		Computed: Extremely High / High / Medium / Low / Very Low
overall_risk_score	DECIMAL(5,2)		Computed composite risk score (1-5 scale)
branch_head_certified	BOOLEAN	DEFAULT false	BH Certificate signed off
branch_head_certified_at	TIMESTAMP		When BH signed the certificate

report_generated_at	TIMESTAMP		When final report was generated
notes	TEXT		General audit engagement notes

audit_team_members

Maps auditors to audit engagements with their roles.

Column	Type	Constraints	Description
id	UUID	PK	Record identifier
audit_id	UUID	FK → audits.id, NOT NULL	The audit engagement
user_id	UUID	FK → users.id, NOT NULL	The auditor assigned
role	VARCHAR(20)	NOT NULL	lead_auditor / team_member / reviewer
assigned_sections	JSONB		Array of section codes this member is responsible for

audit_sections

Tracks completion status for each of the 13+ standardized audit sections within an engagement.

Column	Type	Constraints	Description
id	UUID	PK	Record identifier
audit_id	UUID	FK → audits.id, NOT NULL	Parent audit engagement
section_code	VARCHAR(30)	NOT NULL	Standardized code: RISK_RATING, SUMMARY, CASH_CHECK, BRANCH_DETAILS, REGISTER, LOANS_ADV_1, LOANS_ADV_2, SMA_NPA, NON_FUND, HOUSEKEEPING, STAFF_MATTERS, FIXED_ASSETS, OTHER_OBS
section_name	VARCHAR(100)	NOT NULL	Display name matching Excel sheet tab
status	VARCHAR(20)	DEFAULT 'not_started'	not_started / in_progress / completed / reviewed
assigned_to	UUID	FK → users.id	Team member responsible for this section

section_data	JSONB	NOT NULL DEFAULT '{}'	Structured section data (format varies by section type)
completion_pct	INTEGER	DEFAULT 0	0-100 progress percentage
reviewed_by	UUID	FK → users.id	Lead auditor who reviewed this section
reviewed_at	TIMESTAMP		When section was reviewed

audit_observations

Individual audit findings/observations recorded across all sections. This is the core data table driving risk ratings and compliance tracking.

Column	Type	Constraints	Description
id	UUID	PK	Unique observation identifier
audit_id	UUID	FK → audits.id, NOT NULL	Parent audit engagement
section_code	VARCHAR(30)	NOT NULL	Which audit section this observation belongs to
sr_no	INTEGER	NOT NULL	Sequential number within the section
particulars	TEXT	NOT NULL	Category/area being audited (e.g., Cash Retention Limit)
observation	TEXT	NOT NULL	Detailed audit finding/irregularity
risk_category	VARCHAR(20)	NOT NULL DEFAULT 'low'	extremely_high(5) / high(4) / medium(3) / low(2) / very_low(1)
risk_score	INTEGER	CHECK(1-5)	Numeric risk score: 5=Extremely High, 1=Very Low
pertains_to	VARCHAR(30)		Functional area: finance / operations / legal_recovery / hr / it
branch_comments	TEXT		Initial branch response during audit
is_repeat	BOOLEAN	DEFAULT false	Whether this was flagged in a previous audit
previous_observation_id	UUID	FK → audit_observations.id	Link to prior audit's related observation
regulatory_reference	VARCHAR(100)		RBI circular / bank policy reference if applicable
amount_involved_lakhs	DECIMAL(15,2)		Financial impact of the observation in Lakhs

loan_reviews

Detailed loan account review data from the Loans & Advances sections. Matches the Excel structure precisely.

Column	Type	Constraints	Description
id	UUID	PK	Record identifier

audit_id	UUID	FK → audits.id, NOT NULL	Parent audit
account_number	VARCHAR(20)	NOT NULL	Loan account number (14-digit)
borrower_name	VARCHAR(200)	NOT NULL	Name of borrower / company
product_type	VARCHAR(50)		Loan product (Home Loan, Personal Loan, LAP, etc.)
sanction_date	DATE		Date loan was sanctioned
sanction_amount_lakhs	DECIMAL(15,2)		Sanctioned amount in Lakhs
disbursement_date	DATE		Date of disbursement
disbursement_amount_lakhs	DECIMAL(15,2)		Disbursed amount in Lakhs
outstanding_amount_lakhs	DECIMAL(15,2)		Current outstanding balance
asset_classification	VARCHAR(20)		Standard / SMA-0 / SMA-1 / SMA-2 / NPA
days_past_due	INTEGER	DEFAULT 0	DPD count for overdue classification
audit_observation	TEXT		Auditor's finding on this loan account
risk_category	VARCHAR(20)		Risk rating for this specific account
documentation_status	VARCHAR(20)		complete / incomplete / missing
collateral_verified	BOOLEAN		Whether collateral/security was verified

sma_npa_entries

Asset quality tracking data from the SMA & NPA audit section.

Column	Type	Constraints	Description
id	UUID	PK	Record identifier
audit_id	UUID	FK → audits.id, NOT NULL	Parent audit
category	VARCHAR(20)	NOT NULL	overdue / sma_1 / sma_2 / npa / loss
total_accounts	INTEGER		Number of accounts in this category
total_amount_lakhs	DECIMAL(15,2)		Total amount in Lakhs

audit_observation	TEXT		Auditor's observations on asset quality
risk_category	VARCHAR(20)		Risk rating for this category
pertains_to	VARCHAR(30)		Functional area (typically Legal & Recovery)
legal_notice_issued	BOOLEAN		Whether legal notices have been sent
recovery_action_taken	TEXT		Details of recovery measures
recommended_for_abn	BOOLEAN		Whether file recommended for ABN (write-off)

cash_checks

Cash verification data captured during the surprise cash check at audit commencement.

Column	Type	Constraints	Description
id	UUID	PK	Record identifier
audit_id	UUID	FK → audits.id, NOT NULL, UNIQUE	One cash check per audit
cash_in_hand	DECIMAL(15,2)		Physical cash balance verified
cash_balance_as_per_books	DECIMAL(15,2)		Cash balance per CBS/GL
difference	DECIMAL(15,2)		Computed: cash_in_hand - book_balance
atm_cash_balance	DECIMAL(15,2)		Cash held in ATM cassettes
atm_retention_limit	DECIMAL(15,2)		Authorized ATM retention limit
retention_limit	DECIMAL(15,2)		Branch cash retention limit
excess_cash	DECIMAL(15,2)		Amount exceeding retention limit
soiled_notes	DECIMAL(15,2)		Value of soiled/mutilated notes
cash_check_time	TIMESTAMP		Exact time of surprise cash verification
observations	TEXT		Cash-related audit observations
denomination_data	JSONB		Denomination-wise cash breakdown

RAM and Planning Tables

ram_assessments

Risk Assessment Model (RAM) scores computed for each branch. The 19 parameters from the RBIA Policy are scored to determine audit frequency.

Column	Type	Constraints	Description
id	UUID	PK	Record identifier
branch_id	UUID	FK → branches.id, NOT NULL	Branch being assessed
assessment_year	VARCHAR(10)	NOT NULL	Financial year (e.g., 2024-25)

inherent_risk_scores	JSONB	NOT NULL	19-parameter scores: {deposits: 3, advances: 4, npa: 2, ...}
control_risk_scores	JSONB		Control effectiveness scores per parameter
inherent_risk_total	DECIMAL(5,2)		Weighted total of inherent risk
control_risk_total	DECIMAL(5,2)		Weighted total of control risk
composite_score	DECIMAL(5,2)		Final composite risk score
risk_category	VARCHAR(20)		Derived: high / medium / low
recommended_frequency	INTEGER		Audit frequency in months (12/18/24)
computed_by	UUID	FK → users.id	IAD manager who ran the assessment
approved_by	UUID	FK → users.id	Senior approval for the assessment
approved_at	TIMESTAMP		Approval timestamp

ram_parameters (Reference/Config)

Configuration table defining the 19 RAM parameters, their weights, and scoring criteria as specified in the RBIA Policy.

Column	Type	Constraints	Description
id	UUID	PK	Parameter identifier
param_code	VARCHAR(20)	UNIQUE, NOT NULL	e.g., DEPOSITS, ADVANCES, NPA, FRAUD, FOREX
param_name	VARCHAR(100)	NOT NULL	Full name: Total Deposits, Total Advances, etc.
category	VARCHAR(30)	NOT NULL	inherent_business / inherent_control / control_environment
weight	DECIMAL(5,2)	NOT NULL	Percentage weight in composite score (all must sum to 100)
scoring_criteria	JSONB	NOT NULL	Scoring rules: {1: '<10Cr', 2: '10- 50Cr', 3: '50-100Cr', ...}
max_score	INTEGER	DEFAULT 5	Maximum score (typically 5-point scale)
is_active	BOOLEAN	DEFAULT true	Whether this parameter is currently in use

Compliance Tracking Tables

compliance_items

Tracks each audit observation through the full compliance lifecycle: Branch Response → ZAC Review → ACE Processing → ACB Closure.

Column	Type	Constraints	Description
id	UUID	PK	Compliance item identifier
observation_id	UUID	FK → audit_observations.id, UNIQUE	Source observation
audit_id	UUID	FK → audits.id, NOT NULL	Parent audit for quick querying
branch_id	UUID	FK → branches.id, NOT NULL	Branch responsible for compliance
status	VARCHAR(30)	NOT NULL DEFAULT 'open'	open / branch_responded / zac_reviewed / ace_processed / closed / escalated
branch_response	TEXT		Branch's detailed compliance response
branch_response_date	DATE		When branch submitted response
branch_evidence_files	JSONB		Array of uploaded evidence file references
zac_review_status	VARCHAR(20)		accepted / rejected / needs_more_info
zac_reviewed_by	UUID	FK → users.id	ZAC reviewer
zac_review_date	DATE		ZAC review date
zac_comments	TEXT		ZAC review comments
ace_status	VARCHAR(20)		processed / pending / escalated_to_acb
ace_processed_by	UUID	FK → users.id	ACE officer
ace_processed_date	DATE		ACE processing date
closure_date	DATE		Final closure date
closed_by	UUID	FK → users.id	Who closed the item
days_open	INTEGER		Computed: days from observation to closure/today
is_overdue	BOOLEAN	DEFAULT false	Computed: whether past compliance deadline
escalation_level	INTEGER	DEFAULT 0	0=none, 1=ZAC escalated, 2=ACE escalated, 3=ACB escalated

due_date	DATE		Compliance deadline per policy timelines
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Value Statement / Examination Master Tables

These tables implement the IA Format (Internal Audit Examination Format) which defines the standardized value statements that auditors must examine at every branch. The format is organized by functional areas, each containing numbered examination items with predefined audit checkpoints.

examination_areas

Master table defining the functional areas of the IA examination format. Each area represents a major banking function to be audited.

Column	Type	Constraints	Description
id	UUID	PK	Unique area identifier
area_code	VARCHAR(10)	UNIQUE, NOT NULL	Short code: DEP, REM, CASH, GOV, BILL, CLR, CUST, KYC, IT, HR, ADM, CRD
area_name	VARCHAR(100)	NOT NULL	Full name: Deposits, Remittances, Cash & Currency Chest, etc.
display_order	INTEGER	NOT NULL	Ordering for UI display (1-12)
description	TEXT		Detailed description of what the area covers
risk_weight	DECIMAL(5,2)		Weight of this area in overall risk computation
is_active	BOOLEAN	DEFAULT true	Whether this area is active in current audit template
version	INTEGER	DEFAULT 1	Template version for change tracking

examination_items

Master table of all value statements (examination points) within each functional area. These are the specific checkpoints auditors evaluate during branch visits. Pre-loaded from the IA Format of RBG.

Column	Type	Constraints	Description
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id	UUID	PK	Unique item identifier
area_id	UUID	FK → examination_areas.id, NOT NULL	Parent functional area
item_number	VARCHAR(10)	NOT NULL	Display number within area (e.g., 1, 2, 3 or A1, A2)
particulars	TEXT	NOT NULL	The value statement / examination point text
sub_items	JSONB		Array of sub-points under this item (for multi-part questions)
default_risk_category	VARCHAR(20)		Suggested risk level if finding detected
pertains_to	VARCHAR(30)		Default functional mapping: Finance / Operations / Legal & Recovery / HR / IT
regulatory_reference	VARCHAR(200)		Applicable RBI circular or bank policy reference
guidance_notes	TEXT		Guidance for auditor on how to evaluate this item
display_order	INTEGER	NOT NULL	Ordering within the area
is_active	BOOLEAN	DEFAULT true	Whether item is active in current template
version	INTEGER	DEFAULT 1	Version tracking for template updates

audit_examination_responses

Runtime table capturing auditor responses for each examination item during a specific audit engagement. One row per item per audit.

Column	Type	Constraints	Description
id	UUID	PK	Record identifier
audit_id	UUID	FK → audits.id, NOT NULL	Parent audit engagement
examination_item_id	UUID	FK → examination_items.id, NOT NULL	The value statement being evaluated
area_code	VARCHAR(10)	NOT NULL	Denormalized area code for efficient querying
status	VARCHAR(20)	DEFAULT 'not_examined'	not_examined / compliant /

			non_compliant / not_applicable / partially_compliant
audit_observation	TEXT		Auditor's detailed finding for this item
risk_categorisation	VARCHAR(20)		Auditor-assigned risk: extremely_high / high / medium / low / very_low
pertains_to	VARCHAR(30)		Functional area (may override item default)
branch_compliance	TEXT		Branch response / compliance action taken
evidence_refs	JSONB		Array of evidence file references attached to this item
examined_by	UUID	FK → users.id	Auditor who examined this item
examined_at	TIMESTAMP		When the examination was completed
auto_generated_observation_id	UUID	FK → audit_observations.id	If finding raised, links to the formal observation

Composite unique constraint on (audit_id, examination_item_id) ensures each value statement is examined exactly once per audit.

Supporting Tables

Additional tables required for complete system functionality:

Table	Purpose	Key Columns
users	All system users with role assignments	id, employee_id, name, email, role, zone, is_active
zones	Zonal master for organizational hierarchy	id, zone_code, zone_name, zonal_manager_id
audit_plans	Annual audit plan with branch-frequency mapping	id, financial_year, branch_id, planned_quarter, status
audit_evidence	File attachments linked to observations	id, observation_id, file_name, file_path, file_type, uploaded_by
notifications	In-app and email notification queue	id, user_id, type, title, message, is_read, sent_at
audit_logs	Complete system activity audit trail	id, user_id, action, entity_type, entity_id, old_value, new_value, ip_address
housekeeping_items	150+ standardized checklist items for the Housekeeping section	id, audit_id, sr_no, particular, status, observation, risk_category
register_checks	Register maintenance verification checklist	id, audit_id, register_name, maintained, observation
branch_details	Branch operational data captured during audit	id, audit_id, staff_data, infra_data, operational_metrics (JSONB)
fixed_assets	Fixed asset verification data	id, audit_id, asset_type, book_value, physical_verified, observation
non_fund_facilities	Bank guarantees, LC, non-fund facility reviews	id, audit_id, facility_type, customer, amount, observation, risk
report_templates	Configurable section templates and checklist items	id, section_code, template_data, version, is_active

Entity Relationship Summary

Key relationships in the database:

branches 1:N → **audits** (one branch has many audits over time)

audits 1:N → **audit_sections** (each audit has 13-16 sections)

audits 1:N → **audit_observations** (each audit has 10-30+ findings)

audits 1:N → **loan_reviews** (each audit reviews 70-400 loan accounts)

audit_observations 1:1 → **compliance_items** (every observation must be tracked to closure)

branches 1:N → **ram_assessments** (annual risk assessment per branch)

User Interface Design

The UI follows a responsive dashboard layout optimized for both desktop monitors (audit management) and tablets (field use during branch visits). The design system uses a sidebar navigation pattern with contextual breadcrumbs.

Screen Inventory

Screen	Module	Primary User	Purpose
Login / SSO	Auth	All	Authentication with role-based redirect
Dashboard - IAD Manager	M6	IAD Manager	Audit plan status, team utilization, branch coverage, overdue compliance
Dashboard - Field Auditor	M6	Auditor	My assigned audits, in-progress sections, upcoming schedule
Dashboard - Branch Head	M6	Branch Head	Pending compliance items, audit history, upcoming audits
Dashboard - ACE/ACB	M6	ACE/ACB	Compliance overview, escalations, quarterly summary
Annual Audit Plan	M1	IAD Manager	Plan creation, branch-frequency matrix, team allocation calendar
RAM Computation	M1	IAD Manager	19-parameter scoring interface with auto-computation
Audit Engagement	M2/M3	Lead Auditor	Audit overview, team, progress tracker, section navigation
Section: Risk Rating	M3	Auditor	Branch metadata, audit period, risk summary entry
Section: BH Certificate	M3/M4	Lead Auditor	Branch Head sign-off form, auditor listing
Section: Summary	M3	Lead Auditor	Executive findings table with branch response columns
Section: Cash Check	M3	Auditor	Cash verification form with denomination breakdown and calculations
Section: Branch Details	M3	Auditor	Staff strength, infrastructure, operational data entry
Section: Register	M3	Auditor	Register maintenance checklist with observation fields
Section: Loans & Advances	M3	Auditor	Loan review table with account search, bulk import, risk tagging
Section: SMA & NPA	M3	Auditor	Asset quality metrics form with category-wise entry
Section: Non-Fund	M3	Auditor	Bank guarantees, LC, non-fund facility review
Section: Housekeeping	M3	Auditor	150+ item compliance checklist with status dropdowns

Section: Staff Matters	M3	Auditor	Staff-related audit observations
Section: Fixed Assets	M3	Auditor	Asset verification with serial number tracking
Section: Other Observations	M3	Auditor	Free-form additional findings entry
Examination Checklist	M3	Auditor	12 functional areas with 220+ value statements, status marking, observation entry
Examination Area View	M3	Auditor	Area-specific checklist with compliant/non-compliant status per item
Report Preview	M4	Lead Auditor	Full report preview before generation, section completeness check
Report Export	M4	Lead Auditor	Generate XLSX/PDF in standardized bank format
Compliance Tracker	M5	All	List of observations with status pipeline, filters, search
Compliance Detail	M5	Branch/ZAC	Individual observation compliance workflow with evidence upload
Branch Scorecard	M6	IAD/ACB	Historical audit performance, trend charts, risk evolution
Analytics Explorer	M6	IAD/ACE	Custom queries: by branch, zone, risk category, section, time period
User Management	M7	Admin	CRUD users, role assignment, zone mapping
Branch Master	M7	Admin	Branch data management, category maintenance
Audit Configuration	M7	Admin	Section templates, risk parameters, scoring criteria, notification rules

Key UI Patterns

Audit Execution Interface

The audit execution screen is the primary workspace for field auditors. It uses a tab-based layout matching the Excel sheet tabs that auditors are already familiar with. Key design decisions:

Left Sidebar: Section navigation showing all 13+ audit sections with completion status indicators (not started: gray, in progress: amber, completed: green, reviewed: blue). Clicking a section loads its form.

Main Content Area: Section-specific form with structured data entry. Each section mirrors the corresponding Excel sheet layout but with digital enhancements — dropdowns for risk categories, auto-calculation of totals, inline validation, and auto-save every 30 seconds.

Right Panel (Collapsible): Context panel showing: previous audit's findings for the same section (for comparison), relevant policy references, and observation quick-add form.

Top Bar: Audit metadata (branch name, audit period, visit number), overall progress percentage, and action buttons (Save Draft, Submit for Review, Generate Report).

Observation Entry Pattern

Every audit section uses a consistent observation entry pattern. Each observation row captures:

Field	Input Type	Validation	Source
Sr. No.	Auto-generated	Sequential within section	System
Particulars / Category	Dropdown + free text	Required, from section-specific master list	Template
Audit Observation	Rich text area	Required, min 20 characters	Auditor
Risk Categorization	Dropdown (5 levels)	Required: Extremely High / High / Medium / Low / Very Low	Auditor
Risk Score	Auto-computed	5/4/3/2/1 mapped from category	System
Pertains To	Dropdown	Finance / Operations / Legal & Recovery / HR / IT	Auditor
Branch Comments	Text area	Optional during audit, required for compliance	Branch Head
Amount Involved	Numeric (Lakhs)	Optional, auto-formatted	Auditor
Evidence	File upload	Max 10MB, PDF/Image/Excel	Auditor
Repeat Finding	Toggle + link	If yes, link to previous observation	System/Auditor

Compliance Tracker UI

The compliance tracking interface uses a Kanban-style pipeline view showing observations moving through stages:

Stage	Color	Actions Available	SLA
Open	Red	View finding details, assign to branch	Report submission date
Branch Responded	Orange	Review response, accept/reject, request more info	30 days from audit report
ZAC Reviewed	Yellow	ZAC approval/rejection, escalation	Next ZAC meeting
ACE Processed	Blue	ACE verification, ACB report preparation	Quarterly cycle
Closed	Green	View closure details, audit trail	N/A
Escalated	Dark Red	Escalation reason, escalation level, re-assignment	Per policy escalation timelines

Dashboard Wireframe Specifications

Each role-based dashboard contains the following widget areas:

IAD Manager Dashboard:

Top row: 4 KPI cards (Audits in Progress, Overdue Compliance Items, Branches Pending Audit, Average Risk Score). Middle row: Audit Plan Calendar (Gantt-style showing team assignments across months) + Branch Risk Heat Map (color-coded by zone/risk level). Bottom row: Compliance Aging Chart (bar chart by days open) + Recent Audit Reports table with status badges.

Field Auditor Dashboard:

Top row: 3 KPI cards (My Active Audits, Sections Completed This Week, Pending Reviews). Main area: Current audit engagement card with section-wise progress bars. Below: Upcoming audit assignments with branch details and preparation status.

Branch Head Dashboard:

Top row: 3 KPI cards (Open Compliance Items, Overdue Items, Days Since Last Audit). Main area: Compliance items list sorted by due date with action buttons (Respond, Upload Evidence). History: Previous audit ratings trend chart.

Feature Specifications

M1: Audit Planning Module

RAM Computation Engine

The Risk Assessment Model (RAM) engine implements the 19-parameter scoring system defined in the RBIA Policy to compute branch-level composite risk scores and determine audit frequency.

RAM Parameters (from RBIA Policy):

#	Parameter	Category	Weight	Scoring Basis
1	Total Deposits	Business Risk	8%	Absolute size of deposit portfolio
2	Total Advances	Business Risk	10%	Size and complexity of loan portfolio
3	Gross NPA Ratio	Asset Quality	8%	NPA as percentage of total advances
4	Net NPA Ratio	Asset Quality	5%	Net NPA after provisions
5	SMA-2 Accounts	Asset Quality	5%	Accounts 61-90 days past due
6	Fraud Cases	Operational Risk	7%	Number and value of fraud incidents
7	Staff Accountability	Operational Risk	3%	Pending accountability cases
8	Customer Complaints	Operational Risk	4%	Volume and severity of complaints
9	Previous Audit Rating	Control Environment	8%	Last RBIA rating (1-5 scale)
10	Compliance of Previous Audit	Control Environment	6%	% of previous observations closed
11	Forex Transactions	Business Risk	3%	Volume of forex operations
12	Cash Handling Volume	Operational Risk	4%	Daily cash throughput
13	IT/Cyber Incidents	Operational Risk	3%	Technology-related incidents
14	Regulatory Penalties	Control Environment	5%	RBI penalties or strictures
15	Staff Strength Adequacy	Control Environment	3%	Actual vs sanctioned staff ratio
16	Revenue per Branch	Business Risk	4%	Profitability indicator
17	Off-Balance Sheet Exposure	Business Risk	4%	BG, LC, and contingent liabilities
18	New Product Lines	Business Risk	3%	Complexity from new offerings
19	KYC/AML Compliance	Regulatory Risk	7%	KYC adherence and STR filing

Computation Logic:

Step 1: Each parameter is scored on a 1-5 scale (1 = Very Low Risk, 5 = Extremely High Risk) based on predefined thresholds configured in ram_parameters table.

Step 2: Weighted scores are computed: Parameter Score multiplied by Parameter Weight.

Step 3: Composite Risk Score = Sum of all weighted scores (range: 1.00 to 5.00).

Step 4: Audit frequency is derived from composite score: Score above 3.5 = Annual audit (12 months), Score 2.5 to 3.5 = 18-month cycle, Score below 2.5 = 24-month cycle.

Annual Audit Plan Generator

Based on RAM scores, the system generates the Annual Audit Plan mapping each branch to a target quarter. Features include: automatic scheduling based on last audit date and frequency, team workload balancing across quarters, drag-and-drop re-scheduling with conflict detection, export to Excel format for Board approval, and version tracking for plan revisions.

M3: Audit Execution Module

This is the largest module, providing structured data capture for all 13 standardized audit sections. The design mirrors the existing Excel workbook structure to minimize auditor retraining. At the core of audit execution is the Value Statement Examination Framework derived from the IA Format of RBG, which defines the specific checkpoints auditors must evaluate at every branch.

Value Statement Examination Framework

The IA Format defines 12 functional areas, each containing numbered value statements (examination points) that auditors must evaluate. These serve as the standardized checklist driving the entire audit execution process. The system pre-loads all value statements when an audit is initiated, and auditors mark each as Compliant, Non-Compliant, Partially Compliant, or Not Applicable, with mandatory observation text for any non-compliant findings.

#	Functional Area	Code	Scope of Examination	Approx. Items
1	Deposits	DEP	Verification of deposit accounts, interest rates, TDS compliance, nomination, dormant accounts, unclaimed deposits, DICGC cover, deposit mobilization practices	25-30
2	Remittances	REM	NEFT/RTGS operations, demand drafts, pay orders, collection of outstation cheques, remittance charges, reconciliation of suspense accounts	15-20
3	Cash & Currency Chest	CASH	Cash retention limits, ATM cash management, soiled note remittance, denomination management, currency chest operations (if applicable), surprise cash verification	20-25
4	Government Business	GOV	Government account operations, pension payments, PPF/SSA transactions, revenue collection, stamp duty, GST collections, compliance with government scheme guidelines	15-20
5	Bills	BILL	Bill purchase, bill discounting, documentary credits, bills for collection, usance bills, return of dishonoured bills, follow-up procedures	10-15
6	Clearing	CLR	Clearing operations, CTS compliance, return memo handling, clearing settlement, NACH mandates, ECS processing, dishonour management	10-15
7	Customer Service	CUST	Complaint handling, turnaround time for services, account opening procedures,	15-20

			nomination updates, standing instructions, locker operations, customer education	
8	KYC / AML	KYC	Customer due diligence, periodic KYC updates, suspicious transaction reporting (STR), cash transaction reporting (CTR), risk categorization of customers, FATCA compliance, PEP screening	20-25
9	IT & Cyber Security	IT	CBS operations, password management, user access controls, phishing/vishing awareness, cyber incident reporting, BCP/DR readiness, ATM/POS security, digital banking controls	20-25
10	Human Resources	HR	Staff rotation, leave management, staff accountability, training attendance, disciplinary proceedings, staff meetings, dual control compliance, succession planning	10-15
11	General Administration	ADM	Premises maintenance, record management, safe deposit vault operations, stationery control, insurance coverage, vendor management, fire safety, accessibility compliance	15-20
12	Credit / Advances	CRD	Loan sanctioning process, documentation verification, disbursement controls, security/collateral valuation, NPA identification, provision adequacy, CERSAI registration, recovery actions, restructuring compliance, SARFAESI proceedings	40-50

Total Value Statements: Approximately 220-280 examination points across all 12 functional areas, pre-loaded as a master checklist for each audit engagement.

Examination Workflow per Functional Area

For each functional area, the auditor follows this workflow:

Step 1 — Area Selection: Auditor selects a functional area from the left sidebar. The system displays all pre-loaded value statements for that area in a scrollable checklist format.

Step 2 — Item Examination: For each value statement, the auditor selects a status: Compliant (green check), Non-Compliant (red cross), Partially Compliant (amber warning), Not Applicable (gray dash), or Not Examined (default).

Step 3 — Observation Entry: For Non-Compliant or Partially Compliant items, the system requires a detailed audit observation, risk categorization (Low/Medium/High/Extremely High), and the functional area it pertains to (Finance/Operations/Legal & Recovery/HR/IT).

Step 4 — Evidence Attachment: Auditor can attach photos, documents, or screenshots as evidence for any finding. Files are linked directly to the specific examination item.

Step 5 — Branch Response: Branch Head can enter compliance comments or corrective action for each non-compliant finding directly against the examination item.

Step 6 — Auto-Observation Creation: When an auditor marks an item as Non-Compliant, the system automatically generates a formal audit_observation record linked to the examination response, feeding into the Summary sheet, risk scoring, and compliance tracking pipeline.

Credit / Advances: Deep-Dive Examination

The Credit/Advances area (Code: CRD) is the most extensive functional area, with approximately 40-50 value statements covering the entire credit lifecycle. Given its complexity, this area is further sub-divided into examination groups:

Sub-Group	Focus Area	Key Examination Points
CRD-A: Pre-Sanction	Loan origination and appraisal	Credit proposal documentation, CIBIL/CRIF score verification, income assessment, limit computation methodology, deviation authority compliance
CRD-B: Sanction & Documentation	Approval and legal documentation	Sanction letter terms, security creation, CERSAI registration, title verification, legal opinion, mortgage creation, insurance of assets
CRD-C: Disbursement	Fund release controls	Pre-disbursement conditions, end-use verification, stage-wise disbursement for housing loans, margin maintenance, tie-up verification
CRD-D: Post-Disbursement	Ongoing monitoring	Annual review, stock audit, QIS/QOS submission, collateral re-valuation, loan covenant monitoring, drawing power computation
CRD-E: NPA Management	Stressed asset handling	NPA identification accuracy, provision adequacy, SARFAESI notice compliance, OTS proposals, write-off procedures, recovery suit filing, NCLT references
CRD-F: Special Products	Product-specific checks	Gold loan LTV compliance, education loan moratorium, MSME restructuring, agriculture KCC renewal, SHG-BLP linkage, PMAY subsidies

Section Data Structures

Section	Data Type	Key Fields	Auto-Computations
Risk Rating	Metadata form	Branch name, audit period, team, visit number	Overall risk score from all observations
BH Certificate	Sign-off form	Auditor names, designations, visit dates, BH acknowledgment	Days spent on audit
Summary	Findings table	Sr No, Irregularity description, Branch comments	Count by risk level, section-wise rollup
Cash Check	Verification form	Cash in hand, book balance, ATM balance, retention limit, denominations	Difference, excess over limit, shortage
Branch Details	Data form	Staff strength, infrastructure, CBS details, operational hours	Staff adequacy ratio
Register	Checklist	Register name, maintained (Y/N), observation, recommendation	Compliance percentage
Loans & Advances I	Detail table	Account, borrower, product, sanction, disbursement, outstanding	Portfolio totals, concentration analysis
Loans & Advances II	Review table	Account, borrower, dates, amounts, observation, risk	Risk distribution, amount at risk
SMA & NPA	Summary + detail	Category-wise counts, amounts, actions taken, recommendations	Total stressed assets, migration analysis
Non-Fund	Review table	Facility type, customer, amount, expiry, observation	Exposure totals by facility type
Housekeeping	Checklist (150+)	Particular, status, observation, risk, pertains to	Section-wise compliance score
Staff Matters	Observations	Staff meetings, rotation, training, leave management	Compliance count
Fixed Assets	Verification table	Asset type, description, book value, physical status, observation	Total book value, discrepancy amount
Other Observations	Free form	Sr No, observation, risk, pertains to, recommendation	Additional risk score contribution

Smart Features for Field Auditors

Auto-Save: Form data is saved locally every 30 seconds and synced to server when connected. Works offline with IndexedDB storage, syncing when back online.

Previous Audit Comparison: Side-by-side view of current vs. previous audit findings for the same section. Repeat findings are automatically flagged.

Bulk Loan Import: CSV upload for Loans & Advances section — auditors can export loan data from CBS and import directly, then add observations per account.

Risk Score Calculator: As observations are entered, the system dynamically computes section-wise and overall risk scores using the weighted average methodology defined in RBIA Policy.

Template-Driven Checklists: The IA Format's 12 functional areas and 220+ value statements are pre-loaded as an examination checklist when each audit is initiated. Housekeeping, Register, and other section-specific checklists are also pre-populated from configurable templates. All checklist items can be customized by administrators as policy evolves.

Photo Evidence: Direct camera integration on tablets for capturing evidence photos during branch visit, automatically linked to the relevant observation.

M4: Report Generation Module

Generates the standardized audit report in the exact format currently used (matching the uploaded Excel workbooks). This ensures compatibility with existing bank processes and regulatory submissions.

Excel Report (Primary): Multi-sheet XLSX workbook with 13-16 tabs matching the current structure. Each sheet reproduces the exact column layout, merged headers, and formatting. Formulas are embedded for auto-calculations (e.g., Cash Check totals, Loan summaries).

PDF Report: Formatted PDF version with table of contents, section headers, and page numbers. Designed for printing and archival. Includes the Branch Head Certificate page with signature lines.

Summary Report: Condensed 2-3 page executive summary with key findings, risk rating, and recommendations. Suitable for ZAC/ACB presentations.

Risk Rating Computation

The overall branch audit risk rating is computed as follows, aligned with the RBIA Policy 5-point scale:

Rating	Score Range	Description	Implication
Extremely High Risk	4.1 - 5.0	Critical deficiencies requiring immediate corrective action	Special audit within 3 months, ACB escalation
High Risk	3.1 - 4.0	Significant weaknesses in internal controls	Follow-up audit within 6 months
Medium Risk	2.1 - 3.0	Moderate issues requiring management attention	Standard audit cycle continues
Low Risk	1.1 - 2.0	Minor issues with generally adequate controls	Audit cycle may be extended
Very Low Risk	0.0 - 1.0	Robust controls, minimal findings	Audit cycle may be extended further

The score is computed as: weighted average of all observation risk scores across sections, where weights are determined by the section's relative importance (configurable in admin settings). Repeat findings receive a 1.5x weight multiplier as per RBIA Policy guidance.

M5: Compliance Tracking Module

Implements the full post-audit compliance lifecycle as mandated by the RBIA Policy. Every audit observation automatically creates a compliance tracking item that must progress through defined stages before closure.

Compliance Workflow

Stage	Actor	SLA	System Action
1. Observation Raised	System	On report submission	Auto-create compliance_item, set due_date, notify Branch Head
2. Branch Response	Branch Head	30 days from report date	Branch uploads evidence, enters response. Overdue triggers L1 escalation
3. ZAC Review	Zonal Auditor	Next ZAC meeting	ZAC reviews response: Accept, Reject (with reasons), Request More Info
4. ACE Processing	ACE Officer	Quarterly cycle	ACE verifies closure, prepares ACB report. Unresolved items escalated
5. ACB Reporting	ACB Member	Board meeting cycle	Consolidated compliance status presented. Policy decisions on persistent issues
6. Closure	ACE/Lead Auditor	On satisfactory resolution	Item marked closed with full audit trail. Impacts next RAM score computation

Escalation Rules

Automated escalation based on RBIA Policy timelines:

Trigger	Escalation Level	Action
Branch response overdue by 15 days	Level 1: Reminder	Email reminder to Branch Head with CC to Zonal Manager
Branch response overdue by 30 days	Level 2: ZAC Escalation	Auto-escalate to ZAC, flag on Zonal Manager dashboard
Compliance open > 90 days	Level 3: ACE Escalation	Escalated to ACE department for intervention
Compliance open > 180 days	Level 4: ACB Escalation	Included in ACB exception report for Board attention
Repeat finding (same issue in consecutive audits)	Priority Flag	Marked as persistent issue, 1.5x risk weight in next RAM computation

M6: Analytics & Dashboards Module

Provides real-time audit intelligence through interactive dashboards and reporting. All analytics are role-filtered and support drill-down from summary to individual observations.

Key Analytics Views

Analytics View	Visualization	Data Source	Drill-Down
Branch Risk Heat Map	Color-coded map/grid by zone	ram_assessments + latest audit ratings	Click branch → Branch Scorecard
Audit Plan Progress	Gantt chart with completion status	audit_plans + audits	Click audit → Audit Engagement
Compliance Aging	Stacked bar chart (by days open)	compliance_items grouped by aging bands	Click band → filtered compliance list
Risk Category Distribution	Pie/Donut chart	audit_observations grouped by risk_category	Click category → observation list
Section-wise Findings	Horizontal bar chart	audit_observations grouped by section_code	Click section → section findings
Trend Analysis	Line chart (quarterly)	audits + observations over time	Click point → quarterly details
Top Observations	Ranked table	Most frequent observations across branches	Click observation → occurrences list
Team Productivity	Bar chart per auditor	audits + sections completed per auditor	Click auditor → audit history
NPA Movement	Waterfall chart	sma_npa_entries across audit periods	Click category → account details
Compliance SLA Performance	Gauge + trend	compliance_items vs. due dates	Click zone → zone-level SLA

M8: Enterprise Risk Register & Audit Universe

Beyond branch-level RAM scoring, RBIAS maintains a comprehensive audit universe and enterprise risk register linking all auditable entities with risks, controls, and past audit engagements. This aligns with RBI's RBIA circular which expects the audit function to cover all material risk areas and not be limited to transaction testing at branch level.

Audit Universe

The audit universe captures every entity subject to internal audit, organized by type and linked to the risk register and past audit history.

Entity Type	Examples	Risk Linkage
Branches	Fort Main, Bhandup, Goregaon East, Chembur, etc.	RAM scores, branch risk rating, operational risk profile
Zones / Regions	Western Zone, Central Zone, etc.	Aggregated branch risk, management effectiveness
Products	Home Loan, Gold Loan, KCC, MSME, Personal Loan, etc.	Product-level credit risk, process maturity, regulatory compliance
Processes	Cash Ops, Retail Lending, Treasury, Trade Finance, etc.	Process risk assessment, control effectiveness ratings
IT Systems	CBS, LOS/LMS, Internet Banking, UPI, ATM Switch	Cyber risk, system availability, access control maturity
Projects	Core banking migration, digital lending rollout, etc.	Project risk, timeline, budget, governance
Vendors / Outsourced	IT vendors, BPO partners, cash management agencies	Third-party risk, SLA compliance, regulatory oversight

Enterprise Risk Register

Each auditable entity is linked to a risk register containing structured risk statements with inherent and residual risk assessments.

Column	Type	Description
id	UUID	Risk register entry identifier
entity_type	VARCHAR(30)	branch / zone / product / process / it_system / project / vendor
entity_id	UUID	FK to the specific entity
risk_statement	TEXT	Descriptive risk statement (e.g., 'Inadequate KYC due diligence leading to regulatory penalty')
risk_category	VARCHAR(30)	credit / operational / compliance / strategic / reputational / cyber / liquidity

inherent_risk_score	DECIMAL(5,2)	Pre-control risk rating (1–5 scale)
control_effectiveness	VARCHAR(20)	strong / adequate / weak / non_existent
residual_risk_score	DECIMAL(5,2)	Post-control residual risk (1–5 scale)
risk_owner	UUID	FK → users.id — accountable risk owner
kri_definition	TEXT	Key Risk Indicator definition and threshold
kri_current_value	DECIMAL(10,2)	Latest KRI measured value
kri_breach_status	VARCHAR(20)	within_threshold / warning / breached
linked_controls	JSONB	Array of control_library IDs mapped to this risk
last_audited	DATE	When this risk area was last covered by an audit
next_review_due	DATE	When the risk assessment should be refreshed

Enhanced Risk-Based Plan Inputs

RBI expects risk assessment to consider factors beyond RAM parameters, including past audit results, external inspections, management changes, and environmental factors. The planning module is extended with:

Extended RAM Inputs: Parameters for external audit / regulatory findings, IT and cyber assessments, fraud trends, and market / macro indicators feed into the audit plan prioritization engine alongside the existing 19-parameter RAM model.

Configurable Weighting Layer: IAD managers can tune how much weight RAM scores, external inspection results, thematic risk inputs, and KRI breaches carry in determining audit frequency and scope. Weights are versioned and approval-controlled.

Plan Simulation ('What-If' Analysis): Scenario modeling that shows the impact on audit frequency and schedule when a risk parameter worsens. For example, simulating a 20% increase in NPA at 5 branches shows how many additional audits would be triggered, enabling proactive resource planning.

Audit Engagement → Risk Linkage: Each audit engagement can be linked to one or more auditable entities beyond the branch and to specific risk themes from the risk register (e.g., 'Credit underwriting quality at Branch X' or 'KYC lapses across Western Zone'), enabling thematic audit coverage tracking.

M9: Control Library & Standardized Work Programs

Following global best practices (COSO, IIA Standards, ISO 31000), RBIAS maintains a centralized control library with test procedures and work programs. This shifts the audit approach from being purely observation-focused to one grounded in systematic control testing and effectiveness assessment.

Control Library

Column	Type	Description
id	UUID	Control identifier
control_code	VARCHAR(20)	Unique code (e.g., CRD-CTL-001, CASH-CTL-012)
process_area	VARCHAR(50)	Mapped process: deposits, lending, cash_ops, kyc_aml, it_security, etc.
control_objective	TEXT	What the control aims to achieve
control_description	TEXT	How the control operates — activity, frequency, responsible party
control_type	VARCHAR(20)	preventive / detective / corrective / directive
control_frequency	VARCHAR(20)	per_transaction / daily / weekly / monthly / quarterly / annual
control_owner	VARCHAR(100)	Role or position responsible for operating this control
key_control	BOOLEAN	Whether this is a key control — key controls receive mandatory testing
framework_mapping	JSONB	Mapping to frameworks: COSO, ISO 31000, etc.
linked_risks	JSONB	Array of risk_register IDs this control mitigates
is_active	BOOLEAN	Whether the control is currently active

Test Procedures & Work Programs

Standardized test procedures define how each control should be evaluated. A work program is the set of procedures assigned to a specific audit engagement, with completion and review tracking. When an audit is initiated, the system generates a default work program which the Lead Auditor can customize.

Table	Key Columns	Purpose
test_procedures	id, control_id, procedure_code, description, sample_methodology, expected_evidence, pass_criteria	Standard test templates linked to controls
work_program_items	id, audit_id, test_procedure_id, assigned_to, status, actual_sample_size, result, workpaper_ref, reviewed_by	Runtime work program per audit engagement

Control Effectiveness Analytics

The Analytics module is extended with control effectiveness views: trends across time showing whether controls are improving or degrading, heatmaps by process area and branch, and drill-down from aggregate scores to individual test results.

M10: Continuous Auditing & Data-Driven Testing

Modern internal audit practice emphasizes data analytics and continuous monitoring, testing 100% of significant transactions rather than relying solely on periodic sampling. This module provides data connectors, a rule-based analytics engine, and continuous monitoring dashboards.

Data Integration Layer

Read-only connectors and ETL pipelines bring operational data into RBIAS for automated analysis:

Source System	Data Feed	Frequency	Key Data Points
Core Banking (CBS)	Account and transaction data	Daily	Account master, transaction ledger, interest rates, balance movements
Loan Origination (LOS/LMS)	Loan lifecycle data	Daily	Applications, sanctions, disbursements, repayments, overdue, NPA flags
General Ledger	GL balances and entries	Daily	Suspense balances, reconciliation status, P&L items, off-balance sheet
AML / KYC System	Compliance alerts	Real-time	STR flags, CTR reports, risk categorization changes, PEP matches
ATM / POS Switch	Device and transaction data	Daily	ATM cash levels, failed transactions, reconciliation gaps
HR / Payroll	Staff data	Weekly	Transfers, leave patterns, accountability cases, training records

Rule-Based Analytics Engine

Configurable rule sets that run automatically against ingested data to detect exceptions and anomalies:

Rule Category	Example Rules	Severity
Credit Anomalies	EMI debited but no matching loan in CBS master; Sanction exceeds delegated authority; Incomplete documentation flags	High
Cash Irregularities	Cash exceeding retention limit 3+ days; Large deposits just below CTR threshold (structuring); Denomination imbalance	High
KYC/AML Red Flags	Multiple accounts on same PAN with high turnover; Dormant account large reactivation; PEP without enhanced DD	Critical
NPA Early Warning	SMA-1 not migrated to SMA-2 despite DPD breach; Provision shortfall vs RBI norms; Recovery suits not filed timely	High

Operational Gaps	Aged suspense balances > 30 days; Reversed entries without supervisor approval; Pending inter-branch reconciliation	Medium
Staff-Related	Single user as maker and checker; Excessive override usage; High-value transactions without rotation	Medium

Data Exceptions & Continuous Monitoring

Exceptions detected by rules are stored in a `data_exceptions` table with severity, status tracking, and linkage to audit engagements and formal observations. Continuous monitoring dashboards provide real-time KPIs, early warning lists of branches breaching thresholds, coverage metrics, and exception-to-audit pipeline views.

M11: AI & Advanced Analytics Layer

RBIAS embeds AI and machine learning capabilities to enhance anomaly detection, provide intelligent suggestions during audit execution, and enable predictive risk modeling.

Anomaly Detection

Domain	Model Type	Application
Loan Portfolio	Isolation Forest / Autoencoder	Detect outlier loan accounts deviating from branch or product norms by size, tenure, repayment pattern
Cash Operations	Statistical Process Control + DBSCAN	Identify unusual cash inflow/outflow patterns, denomination anomalies, and timing irregularities
NPA Movement	Time-series anomaly detection	Flag branches where NPA classification timing or provision patterns deviate from regulatory norms
Transaction Monitoring	Clustering (k-means / HDBSCAN)	Surface structuring, round-tripping, or unusual counterparty concentration patterns
Staff Behavior	Behavioral scoring models	Identify staff with unusual override patterns, transaction timing, or access behavior

Text Analytics & Predictive Models

NLP Clustering: Group similar observations across branches and audit periods using text embeddings and topic models, revealing recurring themes.

Auto-Suggest References: When drafting observations, the system suggests applicable RBI circulars, bank policy references, and similar past findings using semantic similarity search.

RAM Category Prediction: Predict probability of a branch moving to higher risk category based on historical patterns of exceptions, fraud incidents, and audit ratings.

Compliance Risk Scoring: Predict which open compliance items are likely to breach SLA deadlines based on response patterns and historical closure times.

Smart Suggestions UI

A context-aware sidebar during observation entry provides: suggested risk category/severity based on text analysis, top 5 similar past findings from prior audits, pre-drafted root cause analysis and recommended corrective actions, and emerging risk theme alerts from the latest audit cycle.

M12: Issue & Action Management

Global best practice treats an ‘issue’ as a reusable concept across all assurance activities — not only internal audit but also regulatory inspections, external audit, and internal control testing. RBIAS extends the compliance tracking model with a unified issue management framework and granular action plan tracking.

Issues Table

Column	Type	Description
id	UUID	Issue identifier
issue_code	VARCHAR(20)	Auto-generated code (e.g., ISS-2026-0042)
source	VARCHAR(30)	internal_audit / regulatory_inspection / external_audit / control_testing / risk_event / self_assessment
title	VARCHAR(200)	Concise issue title
description	TEXT	Detailed issue description
issue_type	VARCHAR(30)	control_deficiency / policy_violation / regulatory_breach / process_gap / system_weakness
severity	VARCHAR(20)	critical / high / medium / low
root_cause	TEXT	Root cause analysis
risk_theme	VARCHAR(50)	Linked risk theme from risk register taxonomy
owner	UUID	FK → users.id — issue owner responsible for remediation
linked_controls	JSONB	Array of control_library IDs related to this issue
linked_compliance_item_id	UUID	FK → compliance_items.id (if sourced from audit)
status	VARCHAR(30)	open / action_in_progress / partially_closed / closed / accepted_risk
due_date	DATE	Target remediation date
residual_risk_accepted	BOOLEAN	Whether residual risk is formally accepted by management

Action Plans

Each issue can have multiple action items, enabling partial closure and milestone tracking:

Column	Type	Description
id	UUID	Action plan item identifier
issue_id	UUID	FK → issues.id
action_description	TEXT	Specific corrective action to be taken

action_owner	UUID	FK → users.id
priority	VARCHAR(20)	immediate / short_term / medium_term / long_term
original_due_date	DATE	Initial target date
revised_due_date	DATE	Revised target (with revision history in JSONB)
milestones	JSONB	Array of intermediate milestones with dates and status
status	VARCHAR(20)	not_started / in_progress / completed / overdue / cancelled
evidence_files	JSONB	Uploaded evidence file references
verified_by	UUID	FK → users.id — auditor who verified closure

This enables the Board/ACB to receive a consolidated view of all open issues regardless of source, with severity-based prioritization and partial closure tracking.

Enhanced Planning & Resource Management

The Audit Planning module is extended with skill-based resource allocation, capacity management, and multi-entity audit support aligned with enterprise-grade audit tools.

Auditor Skills & Capacity

Enhancement	Description
Skills Matrix	Each auditor profile includes domain skills (IT audit, cyber, treasury, retail lending, forex, AML), certifications (CIA, CISA, CFE), and experience levels. Stored as JSONB in extended user profiles.
Utilization Targets	Annual utilization targets (e.g., 80% productive audit days) tracked against actual assignments. Prevents over-allocation and burnout.
Skill-Based Auto-Assignment	System suggests optimal team composition based on branch risk profile and required skills (e.g., high-NPA branch requires credit specialist).
Conflict Detection	Detects utilization breaches, cooling-off periods between same-branch audits, and skill coverage gaps beyond simple date conflicts.

Geography-Aware Scheduling

Travel optimization through geography-aware clustering: the system proposes audit trips that combine nearby branches into efficient travel schedules, reducing cost and time while maximizing coverage. Branches are geocoded and clustered using proximity algorithms.

Multi-Entity Audits

Support for engagements covering multiple branches or entities under a single audit (e.g., thematic audits on 'Gold Loan Processes across Western Zone'). A single engagement links to multiple branch_ids with shared findings, consolidated reporting, and cross-entity comparison analytics.

GRC Integration & Open API Layer

RBIAS provides an open integration layer enabling bi-directional data exchange with the bank's broader GRC ecosystem and enterprise systems, aligning with RBI's expectation that RBIA is a subset of the broader risk management framework.

Integration Points

Integration Point	Direction	Protocol	Purpose
Enterprise ERM System	Bi-directional	REST + Webhooks	Sync risk register entries, KRI values, and risk assessment updates
ITSM / Issue Tracker	Bi-directional	REST API	Push audit issues to IT remediation queues; receive resolution updates
HRMS / IAM	Inbound	SCIM / LDAP / SAML	User provisioning, role sync, staff transfer notifications, SSO
Regulatory Reporting	Outbound	REST / File Export	Audit summaries and compliance data for regulatory submissions
CBS / Core Banking	Inbound (read-only)	DB connector / API	Branch, account, and transaction feeds for continuous auditing
Document Management	Bi-directional	REST / WebDAV	Store and retrieve audit workpapers, evidence, and reports

Shared Taxonomies & Unified Risk View

RBIAS maintains configurable master dictionaries synchronized with enterprise systems: common risk categories, process codes, product codes, control objectives, and organizational hierarchy. A 'Single View of Risk' dashboard aggregates data from multiple sources into one screen per risk theme: showing the risk score, open issues from all assurance sources, pending action plans, recent audit coverage, and current KRI status.

UX & Mobility Enhancements

Building on the responsive web application design, RBIAS includes mobility features, workpaper management, and collaboration tools to fully replace paper-based audit files and email chains.

Mobile Application / PWA

Feature	Platform	Description
Evidence Capture	Mobile / Tablet	Camera integration for photos, video, and voice-to-text notes linked to active audit sections
Quick Checklists	Mobile	Simplified value statement examination with swipe-based status marking optimized for one-handed use
Offline-First Sync	Mobile / Tablet	Full offline capability using service workers and IndexedDB; conflict-free sync on reconnection
Branch Head Sign-Off	Tablet	Digital BH Certificate with signature pad capture and timestamp
Push Notifications	Mobile	Real-time alerts for assignments, deadlines, review requests, and escalation triggers

Electronic Workpaper Management

Workpaper per Test: Each work program item has an associated electronic workpaper with standardized sections: objective, scope, sample selection, test steps, results, conclusion, and reviewer sign-off.

Cross-References & Version History: Internal hyperlinks between workpapers, observations, and evidence files. Full version tracking with diff views.

Export Pack: One-click export of the complete audit file as ZIP/PDF bundle containing all workpapers, evidence, reports, and sign-offs.

Collaboration Features

Inline Comments & Review Notes: Threaded comments with @-mentions on sections, observations, and workpapers. Structured review workflow: Reviewer raises query → Auditor responds → Resolved flag.

Real-Time Presence: See which team members are currently working on which sections during a live audit engagement, preventing duplicate effort.

Multilingual Support: UI localization framework for Hindi and regional languages, supporting multi-lingual audit teams.

Multi-Tenant Architecture

For deployment across multiple UCBs, NBFCs, or small finance banks, RBIAS supports multi-tenancy from day one, enabling SaaS-style commercial delivery without major redesign.

Tenant Isolation & Configuration

Aspect	Approach
Data Isolation	All key tables include tenant_id. PostgreSQL Row-Level Security (RLS) enforces strict isolation at DB layer.
Tenant Configuration	Per-tenant: RAM parameters, scoring weights, templates, workflow rules, report branding, SLAs, escalation timelines.
White-Label Branding	Configurable logo, color palette, report headers/footers, and terminology overrides per bank.
User Management	Tenant-scoped user creation with tenant-admin role. Per-tenant SSO (SAML/OIDC) integration.
Data Residency	Support for tenant-specific schemas or dedicated DB instances for strict data residency requirements.
Billing & Metering	Usage tracking per tenant: active users, audits, storage, API calls. Subscription and usage-based pricing.

Deployment Models

Model	Description	Suitable For
Shared SaaS	Multi-tenant cloud with RLS isolation. Cost-effective, centralized updates.	Small UCBs, cooperative banks
Dedicated Instance	Separate app and DB per bank. Maximum isolation.	Larger banks, strict data sovereignty
On-Premise	Deployed in bank's own data center or private cloud.	Regulatory mandates for on-premise
Hybrid	Cloud platform with on-premise data processing via secure connectors.	Cloud convenience + on-premise security

M13: Quality Management for Internal Audit Function

RBI and IIA both mandate periodic quality assessments of the internal audit function itself. RBIAS includes a Quality Assessment module for ongoing internal assessments and periodic external quality reviews.

Quality Assessment Framework

Component	Description
Self-Assessment Questionnaires	Standardized questionnaires mapped to IIA Standards and RBI RBIA guidance. Scored against maturity criteria.
Gap Analysis	Automated gap identification. Gaps converted to issues in the Issue Management module for tracking.
External QA Support	Template for external quality reviews. Findings imported and tracked alongside self-assessment gaps.
Remediation Tracking	All quality gaps feed into issues and action_plans framework for consistent follow-up and closure.

Internal Audit Effectiveness KPIs

KPI	Measurement	Target
Plan Completion Rate	% of annual audit plan completed on schedule	90–100%
Avg Audit Cycle Time	Days from audit start to final report submission	15–25 days
High-Severity Resolution	% of critical/high issues resolved within SLA	> 85%
Repeat Finding Rate	% of observations that are repeats from prior audits	< 15%
Compliance Closure Rate	% of compliance items closed within policy timelines	> 80%
Stakeholder Satisfaction	Annual survey score from auditees and management (1–5)	> 3.5
Auditor Utilization	% of productive audit days vs. available days	75–85%
Data Analytics Coverage	% of key transactions covered by continuous monitoring	> 60%
Workpaper Quality Score	Average reviewer acceptance rate without rework	> 90%
Training Hours/Auditor	Annual professional development hours	> 40 hrs

These KPIs are tracked in the Analytics module and presented in a dedicated 'Audit Function Health' dashboard available to IAD management and ACB members.

UCB & RBI Regulatory Alignment Framework

RBIAS is designed as a regulatory-aligned Internal Audit & Risk Assurance platform specifically tailored for Urban Cooperative Banks (UCBs). The following sections detail how the system implements and supports RBI's Master Circular on Inspection & Audit Systems in UCBs, IRAC norms, capital adequacy requirements, investment and treasury controls, and inter-bank exposure limits. Together, these capabilities ensure that a UCB using RBIAS can demonstrate full compliance and strong governance during RBI inspections.

M14: Unified Audit Universe & Calendar

RBI expects UCBs to maintain a robust framework of internal inspection, concurrent audit, and EDP/IS audit alongside RBIA. RBIAS extends the audit planning module to cover all mandated audit types in a single calendar, each tagged with the required periodicity and audit characteristics.

Audit Type	RBI Requirement	RBIAS Implementation
Branch RBIA	Risk-based frequency (12/18/24 months) per RAM scores	Core module M1–M4. RAM-driven planning with auto-scheduling
Branch Internal Inspection	At least once in 12 months; surprise character mandatory	Separate scope type with surprise scheduling. Pre-loaded RBI-aligned inspection checklists
Concurrent Audit	Mandatory for large/problem branches, treasury, sensitive areas. Daily/weekly checks	Dedicated concurrent audit workbench (see M15). Scope templates for cash, investments, advances, off-BS items
EDP / IS Audit	Dedicated EDP audit cell; review of CBS, channels, access controls, DR, vendor management	IS/EDP audit module (see M18) with application inventory, IS-specific checklists, and vendor risk tracking
RBI Inspection Follow-up	Systematic tracking of RBI observation paras and ATR submission	Regulatory observation hub (see M19) with para-to-issue mapping and ATR workflow
Statutory Audit Follow-up	Track qualifications, emphasis of matter, and remediation	Statutory observations imported into unified issue register with action plans

Audit Universe Entity Table

Column	Type	Description
id	UUID	Entity identifier
entity_type	VARCHAR(30)	branch / ho_department / process / channel / vendor
entity_name	VARCHAR(100)	Name (e.g., 'Treasury Department', 'Internet Banking Channel')
branch_id	UUID	FK → branches.id (null for non-branch entities)
risk_score	DECIMAL(5,2)	Composite risk score from RAM / concurrent issues / IRAC findings
last_audit_date	DATE	Date of last completed audit/inspection
last_audit_rating	VARCHAR(20)	Rating from last engagement
required_frequency_months	INTEGER	RBI-mandated or risk-derived periodicity
audit_scope_types	JSONB	Applicable types: [rbia, internal_inspection, concurrent, is_audit]
surprise_required	BOOLEAN	Whether RBI mandates surprise character for this entity

status	VARCHAR(20)	active / inactive / merged
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M15: Concurrent Audit Module

RBI requires concurrent audit as an early-warning system covering large/problem branches, treasury, investments, and sensitive areas. RBIAS implements concurrent audit as a first-class module with its own workbench, scope templates, and escalation mechanisms.

Concurrent Audit Scope Templates

Pre-loaded RBI-aligned scope templates with coverage guidance:

Area	Coverage Level	Key Checks
Cash & Vault	100% of high-value transactions	Retention limit compliance, denomination management, vault access controls, ATM reconciliation
Investments / Treasury	100% of trades	SGL/CSGL reconciliation, broker compliance (5% cap), non-SLR limits, HTM/HFT/AFS classification
Advances (Large/Problem)	100% of specified accounts	Sanction terms adherence, drawing power, end-use, NPA identification accuracy, IRAC compliance
Deposits	10–25% sampling	Interest rate compliance, TDS, nomination, dormant account procedures, unclaimed deposits
Housekeeping	Key items check	Inter-branch/inter-bank reconciliation, suspense/sundry ageing, clearing differences
Off-Balance Sheet	100% of new items	LC/BG issuance compliance, limits, documentation, expiry monitoring
KYC/AML	Risk-based sample	Customer due diligence, periodic KYC updates, STR/CTR filing, PEP screening
EDP Controls	Per CBS change cycle	Password practices, access logs, CBS parameter changes, night batch verification

Concurrent Audit Workbench

Rapid Entry UI: Streamlined interface for concurrent auditors (who may be external CA firms) to record daily/weekly observations under RBI-defined heads. Quick status marking (satisfactory / observation raised / serious irregularity) with mandatory detail for observations.

Serious Irregularity Escalation: When a concurrent auditor marks an observation as ‘serious irregularity’, the system immediately notifies HO/IAD management and flags it on the ACB dashboard, as RBI requires immediate escalation of serious issues.

De-duplication with RBIA: Findings raised in concurrent audit automatically surface during RBIA planning for the same branch. Analytics identify branches with persistent concurrent audit issues for higher RAM risk scoring and more intensive RBIA coverage.

M16: IRAC & Provisioning Engine

RBI's Master Circular on Income Recognition, Asset Classification and Provisioning (IRAC) for UCBs is detailed and non-negotiable. RBIAS includes an IRAC computation engine that cross-checks the bank's own NPA classification and provisioning against regulatory norms.

IRAC Computation Pipeline

Step 1 — Data Ingestion: CBS loan data is imported via file upload (CSV/Excel) or API connector into a `loan_raw_imports` staging table. Key fields: account number, borrower, product type, sanction amount, outstanding balance, overdue days (DPD), current classification, current provision amount.

Step 2 — NPA Identification: The engine applies RBI IRAC norms to recompute classification: Standard, Sub-standard, Doubtful (D1/D2/D3 by age buckets), Loss. Special rules for CC/OD accounts (out-of-order criteria), agricultural loans (crop season norms), and restructured assets.

Step 3 — Provision Computation: Required provisioning is calculated per asset class, secured/unsecured status, and age. Results are compared against the bank's actual provisions.

Step 4 — Exception Generation: Mismatches (under-provisioning, mis-classification, delayed NPA recognition) are automatically generated as audit exceptions linked to the branch RBIA and the unified issue register, visible to ACB.

Table	Key Columns	Purpose
<code>loan_raw_imports</code>	<code>id</code> , <code>audit_id</code> , <code>batch_id</code> , <code>account_no</code> , <code>borrower</code> , <code>product</code> , <code>outstanding</code> , <code>dpd</code> , <code>bank_classification</code> , <code>bank_provision</code>	Staging table for CBS loan data imports
<code>irac_computed_positions</code>	<code>id</code> , <code>import_id</code> , <code>account_no</code> , <code>computed_classification</code> , <code>computed_provision</code> , <code>bank_classification</code> , <code>bank_provision</code> , <code>deviation_type</code> , <code>deviation_amount</code>	Engine output with per-account IRAC recomputation and deviation flags

M17: Investment & Treasury Audit Controls

RBI has strong expectations on investment management in UCBs, including SGL/CSGL reconciliation, broker intermediation controls, non-SLR investment caps, and quarterly certification requirements.

Control Area	RBI Requirement	RBIAS Implementation
SGL/CSGL Reconciliation	Regular reconciliation of government securities holdings	Checklist items in concurrent and periodic audits verifying SGL/CSGL match with bank records
Broker Compliance	Max 5% of annual trade volume per broker; no broker intermediation in inter-bank G-sec trades	Analytics engine calculates per-broker volume percentage and flags breaches with reason-capture for ACB review
Non-SLR Investments	Cap at 10% of total deposits; rating and prudential criteria	Dashboard tracking non-SLR exposure vs cap with drill-down to individual holdings and rating compliance
HTM/HFT/AFS Classification	Proper classification per RBI investment norms	Audit checklist verifying classification correctness and compliance with shifting limits
Quarterly Certification	Concurrent/internal auditor certifies that reported investments are actually held	Workflow for auditor certification with sign-off, upload, and 'sent to RO' status tracking

The HO treasury is treated as an auditable entity in the audit universe, with dedicated concurrent audit and periodic IS audit coverage.

M18: EDP / IS Audit Module

RBI explicitly requires dedicated EDP/IS audit capability in UCBs, covering all critical applications, vendor management, and technology controls.

Application Inventory

Column	Type	Description
id	UUID	Application identifier
app_name	VARCHAR(100)	CBS, Internet Banking, Mobile Banking, ATM Switch, Card Host, UPI, LOS/LMS, etc.
vendor	VARCHAR(100)	Technology vendor name
version	VARCHAR(30)	Current version deployed
hosting	VARCHAR(30)	on_premise / cloud / managed_service
sla_ref	VARCHAR(50)	Reference to vendor SLA document
visitorial_rights	BOOLEAN	Whether contract includes RBI visitorial rights clause
last_is_audit_date	DATE	Date of last IS/EDP audit
criticality	VARCHAR(20)	critical / high / medium / low
dr_tested_date	DATE	Last DR drill date for this application

IS Audit Checklists

Pre-loaded RBI-aligned IS audit checklists covering: user access management and segregation of duties, change management and release controls, backup and disaster recovery drills, security controls (firewall, encryption, endpoint protection), parallel run and cut-over processes for system changes, CBS parameter audit (interest rates, product masters, user privileges), and outsourced processing arrangement oversight including vendor SLA compliance.

For branches with heavy digital channel usage, RBIA automatically includes IS control verification items (password practices, CBS access log sampling) to complement the HO-level IS audit.

M19: Regulatory Observation & Follow-up Hub

RBI expects systematic, timely follow-up of all assurance source observations with ACB oversight. RBIAS provides a unified hub for tracking observations from RBI inspections, statutory audits, concurrent audits, and internal inspections.

Column	Type	Description
id	UUID	Regulatory observation identifier
source	VARCHAR(30)	rbi_inspection / statutory_audit / concurrent_audit / nabard_inspection / other
reference_no	VARCHAR(50)	RBI letter number, audit report reference, etc.
para_no	VARCHAR(20)	Para/observation number within the source report
observation_text	TEXT	Full text of the regulatory observation
severity	VARCHAR(20)	critical / major / minor / procedural
date_received	DATE	Date observation was received by the bank
response_due_date	DATE	Deadline for ATR/response submission
atr_status	VARCHAR(20)	draft / submitted / rbi_accepted / further_info_required
atr_text	TEXT	Action Taken Report content
linked_issue_ids	JSONB	Array of issue IDs mapped to this observation for internal tracking

Each RBI inspection para is mapped to one or more internal issues with action plans, owners, and deadlines. The system tracks ATR preparation, submission to Regional Office, and closure evidence. ACB receives a consolidated view of all regulatory observations and their rectification status.

Housekeeping Risk Metrics

RBI repeatedly flags poor housekeeping as a systemic risk in UCBs. RBIAS maintains branch-level housekeeping KPIs that feed into RAM scoring and trigger more intensive audit coverage:

Metric	Source	Impact
Inter-branch unreconciled entries (count & age)	CBS reconciliation data / audit verification	High unreconciled counts increase RAM score; oldest entry age triggers escalation
Inter-bank unreconciled entries	Treasury/accounts reconciliation	Feeds inter-bank risk dashboard; aged items flagged to ACB
Suspense/sundry balance & age	GL data / audit verification	Large/aged suspense balances increase operational risk score
Clearing differences count & resolution time	Clearing department data	Persistent differences flagged in concurrent and RBIA reports
Off-BS register completeness (LC/BG)	Concurrent audit / RBIA checks	Incomplete registers flagged as control deficiency

These metrics are stored in a `housekeeping_metrics` table (`branch_id`, `as_of_date`, metric values) updated during audits and optionally via data feeds. A dedicated 'Housekeeping Health' dashboard is available for CEO and ACB.

Inter-bank Exposure Monitoring

RBI caps inter-bank exposures for UCBs and requires Board oversight:

Limit	RBI Norm	RBIAS Monitoring
Total inter-bank placements	$\leq 20\%$ of total deposits	Real-time dashboard showing current placement vs cap; breach alerts to treasury and ACB
Per-bank exposure	$\leq 5\%$ of total deposits per counterparty	Per-counterparty exposure tracker with breach flags
Exposures to weak/AID banks	Phased provisioning and exposure reduction	Separate tracking of exposures to banks under RBI directions with provisioning compliance checks
UCB-to-UCB deposits	Specific RBI restrictions apply	Tagged separately in exposure dashboard with applicable limit calculations

Audit checklists for HO/treasury audits include standardized tests checking compliance with all inter-bank limits. Any breaches are directly linked to the issue register and ACB dashboard.

M20: Governance & Board Compliance Module

RBI lays down explicit expectations for Board, ACB, and (for larger UCBs) Risk Management Committee governance. RBIAS supports these governance requirements with:

ACB Workspace & Agenda Builder

A dedicated ACB view that consolidates:

Audit Status Overview: Status of all internal inspections, concurrent audits, EDP audits, and RBIA engagements in one dashboard.

Unrectified Irregularities: Ageing analysis of all open issues across assurance sources, with drill-down by severity and source.

ACB Agenda Builder: Auto-generate quarterly ACB meeting packs containing: summary of serious irregularities, fraud/vigilance cases, long-outstanding inter-branch/inter-bank items, compliance status on RBI/statutory/concurrent observations, and IRAC deviation summaries.

Board Review Calendar

Maintains RBI's suggested list of items to be placed before Board/ACB with required frequency:

Review Item	Frequency	Status Tracking
Asset quality and NPA review	Quarterly	Auto-populated from IRAC engine; presentation and minutes upload
Investment portfolio review	Quarterly	Data from treasury audit module; non-SLR compliance status
Audit follow-up status	Quarterly	Consolidated from issue register across all sources
Risk management review	Quarterly	RAM scores, KRI dashboard, inter-bank exposure status
Policy review status	Annual / As due	Policy library flags overdue reviews; minutes linkage
IT/Cyber security review	Half-yearly	IS audit findings, incident reports, DR drill status
Fraud cases and vigilance	As they arise + Quarterly summary	Linked to issue register with fraud case tracking

Policy Library & Review Tracker

Column	Type	Description
id	UUID	Policy identifier
policy_name	VARCHAR(200)	Credit Policy, Investment Policy, IRAC Policy, Internal Audit Policy, IS/IT Policy, ALM Policy, etc.
category	VARCHAR(30)	credit / investment / audit / risk / it / compliance / hr / operations
approval_date	DATE	Date of last Board/ACB approval
last_review_date	DATE	Date of last review
next_review_due	DATE	System-calculated or manually set next review date
owner	UUID	FK → users.id — department head responsible
file_ref	VARCHAR(200)	Reference to uploaded policy document
board_minutes_ref	VARCHAR(100)	Reference to Board/ACB meeting where policy was approved

RBIA and ACB views flag policies overdue for review. The system maintains version history so auditors can verify which policy version was in effect during a specific audit period.

Committee Governance Metadata

RBIAS tracks Board committee composition and meeting records to demonstrate RBI-compliant governance:

Table	Key Columns	Purpose
board_committees	id, name (ACB / RMC / ALCO / Credit Committee), composition_requirements, active	Master list of governance committees with RBI-mandated composition rules
committee_members	committee_id, user_id, role (chair/member/secretary), appointed_date, qualification_notes	Tracks membership including RBI requirement for at least one CA/finance expert on ACB
committee_meetings	committee_id, meeting_date, agenda_ref, minutes_ref, reports_presented (JSONB)	Meeting records linked to RBIAS-generated reports and issues reviewed

RBI Inspection Support Pack

A one-click report generation feature producing a comprehensive documentation pack for RBI inspection teams:

Report Component	Data Source	Content
Annual Internal Audit Plan Summary	annual_plans + audit_universe_entities	Planned vs completed audits by type, with deviation explanations
Serious & Long-Pending Findings	issues (severity = critical/high, age > 90 days)	All high-severity open issues with ageing, owner, and remediation status
IRAC & Provisioning Deviations	irac_computed_positions	Branch-wise summary of classification and provisioning mismatches
Investment & Inter-bank Compliance	Treasury audit module + inter-bank exposure	Non-SLR cap compliance, broker volume analysis, inter-bank limit adherence
Concurrent Audit Summary	concurrent audit observations	Trend of concurrent audit findings by category and severity
Follow-up on Previous RBI Observations	regulatory_observations + issues	Para-by-para status of previous RBI inspection findings with ATR evidence
Housekeeping Health Summary	housekeeping_metrics	Branch-wise reconciliation status, suspense/sundry ageing, clearing differences
ACB Meeting Records	committee_meetings	Dates, attendance, key decisions, reports reviewed
Policy Review Status	policy_documents	All key policies with last review date and next due date

Risk Management MIS for RMC / Board

Pre-configured risk dashboards aligned to RBI's risk management expectations for UCBs:

Dashboard	Key Metrics	RBI Alignment
Capital Adequacy	CRAR vs 9% Tier-1, 12% overall; Net Worth progression vs phase-in requirements	Master Circular on Capital Adequacy
Asset Quality	Gross NPA %, Net NPA %, SMA pipeline, sector/borrower concentration	IRAC norms; Large exposure framework
Liquidity Risk	SLR position, LCR (if applicable), deposit concentration, inter-bank dependence	ALM guidelines for UCBs
Investment Risk	Portfolio composition (SLR/non-SLR), MTM impact, broker exposure, duration risk	Investment guidelines for UCBs
Operational Risk	Fraud cases, housekeeping health, staff-related issues, IT incident count	Operational risk framework
Credit Concentration	Top 10/20 borrowers exposure, sector-wise concentration, group exposure vs limits	Exposure norms for UCBs

These dashboards can ingest data from the bank's MIS via read-only feeds into a `regulatory_metrics` table, or be populated during the audit cycle. The Risk Management Committee (for UCBs with assets $\geq$ Rs 5000 crore) and the Board receive these as pre-configured report packs.

Non-Functional Requirements

Category	Requirement	Specification
Performance	Page Load Time	All pages load within 2 seconds on standard broadband connection
Performance	Form Auto-Save	Local save within 500ms, server sync within 3 seconds
Performance	Report Generation	Excel report generated within 10 seconds for standard audit (400 loan records)
Performance	Dashboard Refresh	Analytics queries execute within 5 seconds for 3-year data range
Scalability	Concurrent Users	Support 100+ simultaneous users (auditors, branch heads, management)
Scalability	Data Volume	Handle 500+ branches, 2000+ audits/year, 50,000+ observations
Security	Authentication	SSO via SAML/OIDC integrated with bank's IAM; JWT tokens with 15-min access, 7-day refresh; HTTPS only; MFA for sensitive operations
Security	Authorization	Row-level security (RLS) in PostgreSQL; RBAC middleware; auditors see only assigned audits; branches see only their data; just-in-time provisioning via SCIM/LDAP
Security	Data Encryption	AES-256 for data at rest; TLS 1.3 in transit; field-level encryption for PAN, Aadhaar, account numbers; configurable data masking by role
Security	Audit Trail	Immutable audit log for all data modifications with user, timestamp, IP address; shipped to centralized SIEM
Security	Password Policy	Minimum 12 characters, complexity requirements, 90-day rotation (or SSO-delegated)
Security	Change Management	Versioned configuration for RAM parameters, risk scoring, templates, reports; approval workflow for critical settings with full history
Availability	Uptime SLA	99.5% availability during business hours (8 AM - 8 PM IST)
Availability	Backup	Daily automated backups with 30-day retention, tested quarterly
Availability	Disaster Recovery	RPO: 1 hour, RTO: 4 hours; active-passive or active-active deployment options across data centers
Observability	Monitoring	Centralized logs, metrics via Prometheus/Grafana, alert thresholds for error rates and performance; health dashboards
Usability	Offline Support	Audit execution forms work offline with auto-sync when connected
Usability	Browser Support	Chrome 90+, Edge 90+, Firefox 90+, Safari 15+
Usability	Responsive Design	Full functionality on tablets (1024px+), read-only on mobile (768px+)
Compliance	Data Retention	Audit data retained for 8 years per RBI guidelines; concurrent audit records for 10 years; IRAC computation logs retained for regulatory examination periods

Compliance	Export Capability	All data exportable in CSV/Excel/PDF format for regulatory submissions; RBI-friendly report packs (inspection support pack, ATR, concurrent audit summaries) available as one-click exports
Compliance	Regulatory Auditability	Complete audit trail for every observation, classification, and provisioning computation; immutable logs for IRAC engine inputs/outputs; concurrent audit timestamp integrity for RBI examination
Compliance	Function-Level RBAC	Granular permission model aligned to RBI-expected segregation: field auditors, concurrent auditors, IS/EDP auditors, ACB members, and RMC members each see only their authorized modules and data
Compliance	RBI Reporting	Pre-configured report templates for ATR submissions, concurrent audit summaries, IRAC deviation reports, inspection support packs, and regulatory observation follow-up status

Implementation Roadmap

The implementation follows an agile methodology with 2-week sprints, organized into nine phases. Phases 1–4 (6 months) deliver the core RBIA platform; Phases 5–7 (additional 6–9 months) deliver advanced GRC modules; Phases 8–9 (additional 4–6 months) deliver UCB-specific regulatory alignment modules including concurrent audit, IRAC engine, treasury controls, EDP/IS audit, governance dashboards, and the RBI inspection support pack.

Phase 1: Foundation (Weeks 1-6)

Sprint	Deliverables	Priority
Sprint 1-2	Project setup: database schema, Prisma models, Express API skeleton, React scaffold, authentication system with JWT and RBAC middleware	Critical
Sprint 3	Branch Master module, User Management, Zone hierarchy, base dashboard layout with role-based routing	Critical

Phase 2: Audit Execution (Weeks 7-14)

Sprint	Deliverables	Priority
Sprint 4-5	Audit Execution Module: section framework, Cash Check, Branch Details, Register sections with form validation and auto-save	Critical
Sprint 6-7	Loans & Advances sections (I and II) with bulk import, SMA & NPA section, observation entry pattern with risk categorization	Critical
Sprint 8	Remaining sections (Non-Fund, Housekeeping, Staff Matters, Fixed Assets, Other Observations), section review workflow	High

Phase 3: Reporting & Compliance (Weeks 15-20)

Sprint	Deliverables	Priority
Sprint 9-10	Report Generation Module: Excel export matching existing bank format, PDF report, Summary report, BH Certificate	Critical
Sprint 11-12	Compliance Tracking Module: full lifecycle workflow, Branch response portal, ZAC/ACE/ACB review screens, escalation engine	Critical

Phase 4: Planning & Analytics (Weeks 21-26)

Sprint	Deliverables	Priority
Sprint 13-14	RAM Computation Engine with 19 parameters, Annual Audit Plan generator, plan approval workflow	High

Sprint 15-16	Analytics dashboards, Branch Scorecards, trend analysis, management reporting, notification system	High
Sprint 17	UAT, performance optimization, offline support, data migration from existing Excel reports, documentation	Critical

Phase 5: Advanced Analytics & Continuous Auditing (Weeks 27-36)

Sprint	Deliverables	Priority
Sprint 18-19	Data Integration Layer: CBS, LOS/LMS, GL connectors with ETL pipelines; staging tables for key domains; data quality validation framework	High
Sprint 20-21	Rule-Based Analytics Engine: configurable rule sets for credit, cash, KYC/AML, NPA, operational, and staff-related domains; data_exceptions table and management UI	High
Sprint 22-23	Continuous Monitoring Dashboards: real-time exception KPIs, early warning lists, coverage metrics; AI anomaly detection models (isolation forest, clustering); Smart Suggestions sidebar for observation entry	Medium

Phase 6: GRC Integration & Issue Management 2.0 (Weeks 37-46)

Sprint	Deliverables	Priority
Sprint 24-25	Enterprise Risk Register & Audit Universe (M8): auditable entities, risk statements, KRIs, risk-plan linkage, what-if simulation	High
Sprint 26-27	Control Library & Work Programs (M9): control_library, test_procedures, work_program_items tables; standard work program generation per audit; control effectiveness analytics	High
Sprint 28-29	Issue & Action Management (M12): unified issues table, action_plans, multi-source ingestion (regulatory, external audit, self-assessment); consolidated Board reporting views	High
Sprint 30	GRC Integration APIs: REST/webhook connectors for ERM, ITSM, HRMS/IAM; shared taxonomy management; unified risk view dashboard	Medium

Phase 7: Mobility, UX & Productization (Weeks 47-52+)

Sprint	Deliverables	Priority
Sprint 31-32	Mobile PWA: evidence capture (photo/video/voice), quick checklists, offline-first sync, push notifications; electronic workpaper management system	Medium
Sprint 33-34	Collaboration features: inline comments, review notes workflow, real-time presence; Quality Management module (M13): self-assessments, KPIs, external QA support	Medium

Sprint 35-36	Multi-tenant architecture: tenant_id RLS, per-tenant config, white-label branding, deployment model support; NLP text analytics and predictive risk models	Medium
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Phase 8: UCB RBIA Core & Governance (Weeks 53-62)

This phase delivers the foundational UCB regulatory alignment modules, establishing the audit universe, concurrent audit workbench, and governance framework mandated by RBI.

Sprint	Deliverables	Priority
Sprint 37-38	Unified Audit Universe & Calendar (M14): auditable entity registry (branches, HO depts, channels, vendors), risk-tagged calendar with audit-type periodicity (RBIA, concurrent, IS/EDP, statutory), Board-approved plan generation	Critical
Sprint 39-40	Concurrent Audit Module (M15): scope templates for cash, investments, advances, off-BS items; rapid observation entry workbench; serious irregularity escalation with automatic routing; de-duplication logic against RBIA findings	Critical
Sprint 41-42	Governance & Board Compliance (M20): ACB agenda builder with auto-populated packs (serious irregularities, frauds, compliance status, IRAC deviations); Board review calendar; policy document library with version control and periodic review reminders	Critical
Sprint 43	Regulatory Observation Follow-up Hub (M19): unified tracker for RBI inspection, statutory audit, concurrent audit, and internal audit observations; ATR workflow with drafting, review, and submission tracking; ageing analysis and escalation triggers	High

Phase 9: IRAC, Treasury, EDP & Risk Analytics (Weeks 63-72)

This phase delivers the specialized computation engines and regulatory control modules that enable a UCB to demonstrate full IRAC compliance, investment governance, technology audit readiness, and risk management maturity during RBI inspections.

Sprint	Deliverables	Priority
Sprint 44-45	IRAC & Provisioning Engine (M16): CBS data ingestion pipeline, NPA recomputation engine (sub-standard/doubtful D1-D3/loss classification), provisioning cross-check, deviation detection with auto-exception generation, branch-wise deviation dashboard	Critical
Sprint 46-47	Investment & Treasury Audit Controls (M17): SGL/CSGL reconciliation, broker intermediation controls, non-SLR cap monitoring, HTM/AFS/trading portfolio analysis, quarterly auditor certification workflow	High
Sprint 48-49	EDP/IS Audit Module (M18): application inventory with risk scores, IS audit checklist engine (CBS, channels, access, BCP/DR, vendor, change management), technology control evidence collection and gap analysis	High

Sprint 50-51	Risk Management MIS & Inspection Pack: housekeeping KPI dashboards, inter-bank exposure monitoring, 9-component RBI inspection support pack generator, RMC/Board risk dashboards (credit, market, operational, liquidity concentration)	High
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Appendix

A. Glossary

Term	Definition
RBIA	Risk-Based Internal Audit — the methodology where audit scope and frequency are determined by risk assessment
RAM	Risk Assessment Model — the 19-parameter framework for computing branch-level risk scores
IAD	Inspection and Audit Department — the bank's central audit function
ZAC	Zonal Audit Committee — zonal-level committee reviewing audit compliance
ACE	Audit Compliance and Evaluation — department responsible for tracking audit compliance to closure
ACB	Audit Committee of the Board — Board-level committee with oversight of audit function
IA Format	Internal Audit Examination Format — the standardized checklist of 12 functional areas and 220+ value statements defining what must be examined during each branch audit
Value Statement	A specific examination point/checkpoint within a functional area that auditors must evaluate during a branch audit (e.g., 'Whether nomination is obtained in all deposit accounts')
SMA	Special Mention Account — accounts showing early signs of stress (SMA-0: 1-30 DPD, SMA-1: 31-60 DPD, SMA-2: 61-90 DPD)
NPA	Non-Performing Asset — account classified as substandard (90+ DPD), doubtful, or loss
DPD	Days Past Due — number of days a loan payment is overdue
BH Certificate	Branch Head Certificate — formal acknowledgment by branch manager of audit findings
CBS	Core Banking Solution — the bank's centralized banking software system
ABN	Abandonment — write-off recommendation for irrecoverable accounts
KYC/AML	Know Your Customer / Anti-Money Laundering — regulatory compliance requirements
Lac/Lakh	Indian numbering: 1 Lakh = 100,000 (displayed as financial amounts in reports)
GRC	Governance, Risk, and Compliance — integrated framework for managing organizational governance, risk management, and regulatory compliance
KRI	Key Risk Indicator — a metric used to signal increasing risk exposure requiring management attention
COSO	Committee of Sponsoring Organizations — framework for internal control and enterprise risk management
IIA	Institute of Internal Auditors — global professional body setting standards for internal audit practice
ETL	Extract, Transform, Load — data pipeline process for ingesting data from source systems into RBIAS for analytics

RLS	Row-Level Security — PostgreSQL feature enforcing data isolation at the database query level
PWA	Progressive Web Application — web app with native-like mobile capabilities including offline support
UCB	Urban Cooperative Bank — cooperative bank operating in urban/semi-urban areas, regulated by RBI and the respective State Registrar of Cooperative Societies
IRAC	Income Recognition, Asset Classification and Provisioning — RBI norms governing NPA identification, classification (Sub-standard, Doubtful D1/D2/D3, Loss), and mandatory provisioning percentages
CRAR	Capital to Risk-weighted Assets Ratio — capital adequacy measure mandated by RBI for UCBs (minimum 9%)
ATR	Action Taken Report — formal report submitted to RBI detailing compliance actions taken against inspection observations
SGL/CSGL	Subsidiary General Ledger / Constituent SGL — RBI's securities holding and settlement system; UCBs hold government securities via CSGL accounts
RMC	Risk Management Committee — Board-level committee required for UCBs with assets $\geq$ Rs 5000 crore, overseeing credit, market, operational, and liquidity risk
Concurrent Audit	A real-time or near-real-time audit of transactions at branches/HO functions, mandated by RBI for large/problem branches, treasury, and sensitive areas in UCBs
EDP/IS Audit	Electronic Data Processing / Information Systems Audit — technology audit covering CBS, digital channels, access controls, vendor management, and IT governance
SLR	Statutory Liquidity Ratio — minimum percentage of net demand and time liabilities that banks must maintain in liquid assets (government securities, cash, gold)
ALM	Asset-Liability Management — framework for managing risks arising from mismatches between assets and liabilities across time buckets
MTM	Mark-to-Market — revaluation of investment portfolio to current market prices, impacting P&L for trading/AFS categories

B. Reference Documents

1. Risk Based Internal Audit Policy 2018 (RBIA Policy) — the governing policy document defining audit methodology, RAM framework, risk scoring, reporting requirements, and compliance tracking procedures.
2. Internal Audit Format of RBG (IA Format) — the detailed examination checklist defining 12 functional areas and 220+ value statements that auditors must evaluate at each branch. Areas covered: Deposits, Remittances, Cash & Currency Chest, Government Business, Bills, Clearing, Customer Service, KYC/AML, IT & Cyber Security, Human Resources, General Administration, and Credit/Advances.
3. Existing Audit Report Excel Workbooks (11 files analyzed) — actual branch audit reports from Fort, Bhandup, Goregaon, Chembur, Mulund East, Borivli, and Andheri branches, used as the definitive reference for report format and data structure.

4. RBI Guidelines on Risk-Based Internal Audit for Banks — regulatory framework mandating RBIA adoption.
5. Global Internal Audit Best Practices — guidance from IIA Standards, COSO Internal Control Framework, ISO 31000, and leading GRC platforms (AuditBoard, MetricStream, Diligent, Centraleyes) informing the advanced modules (M8–M13) including control library, continuous auditing, AI analytics, and multi-tenant architecture.
6. RBI Master Circular on Inspection & Audit Systems in Urban Cooperative Banks — regulatory framework governing RBIA, concurrent audit, EDP/IS audit, and governance expectations for UCBs, informing modules M14–M20.
7. RBI Master Circular on Income Recognition, Asset Classification and Provisioning (IRAC) for UCBs — norms for NPA identification, classification, and provisioning percentages driving the IRAC engine (M16).
8. RBI Guidelines on Investment by UCBs — SLR/non-SLR limits, SGL/CSGL requirements, broker intermediation controls, and quarterly certification requirements informing investment audit controls (M17).
9. RBI Guidelines on Capital Adequacy (CRAR) for UCBs — minimum 9% CRAR requirement, risk-weight calculation, and capital planning framework referenced in risk management dashboards.
10. RBI Framework on Inter-Bank Exposure Limits for UCBs — single-bank, aggregate, and UCB-to-UCB exposure limits informing the inter-bank exposure monitoring module.

C. Data Migration Strategy

Existing Excel audit reports (11 files) can be migrated into RBIAS using an automated import utility. The migration process involves: parsing each Excel workbook's 13-16 sheets, mapping data to the database schema, handling legacy XLS format conversion, validating imported data against schema constraints, and generating a migration report showing imported records, skipped records, and data quality issues. The bulk import for Loans & Advances sections supports CSV format, which can be extracted from CBS.